Project Report On

A RAG based Biomedical Research Assistant

Submitted in partial fulfillment of the requirements for the award of the degree of

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CERTIFICATE

This is to certify that the project report entitled ***"A RAG based Biomedical Research Assistant"*** *is a bonafide record of the work done by* ***Abel T John(U2203007)***, ***Abhay P (U2203008)***, ***Achyuth Pradeep(U2203015)***, ***Albin Sam Alex(U2203025)***, *submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science And Engineering" during the academic year 2025-2026.*

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Abstract

The rapid increase in the volume of biomedical literature has made it increasingly challenging to efficiently retrieve relevant information from large volumes of scientific documents. The traditional approach to retrieving information using a set of predefined keywords may not yield accurate results due to the large volume of information. This project proposes a system to provide accurate answers to a set of questions in the biomedical domain using the Retrieval Augmented Generation (RAG) pipeline along with Large Language Models (LLMs). The proposed system retrieves relevant scientific papers from the biomedical database and uses a chunking technique to divide the papers into multiple chunks. The most relevant information from the chunks is provided to the language model to generate concise answers to the questions. This approach ensures the accuracy of the answers provided to the users. Experiments are conducted on the EuropePMC dataset, and answer relevancy and faithfulness evaluation metrics are used to show the scores of trustworthiness. Investigations demonstrate a score of 0.78 for answer relevancy and 0.97 for faithfulness.

Contents

List of Abbreviations

LLM- Large Language Models

RAG- Retrieval Augmented Generation

API- Application Programming Interface

BERT- Bidirectional Encoder Representation Transformer

FAISS- Facebook AI Similarity Search

MMR- Maximal Marginal Relevancy

JSON- JavaScript Object Notation

URL- Uniform Resource Locator

DOI- Digital Object Identifier

NLTK- Natural Language Toolkit

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Chapter 1

Introduction

1.1 Background

Biomedical research produces a tremendous volume of scientific literature, including research papers, clinical studies, and medical reports. It is a common occurrence for researchers and medical personnel to require searching through large volumes of scientific literature to obtain relevant and accurate information. Conventional search systems usually employ keyword-based searching, but this is not effective in fully comprehending the meaning of a query. Conventional searching systems require manual reading of various papers to obtain useful information.

Recent advances in Large Language Models (LLMs) have enabled the development of question answering systems that generate responses in natural language. However, these question answering systems sometimes generate false information. Retrieval-Augmented Generation (RAG) is a model that overcomes this limitation by generating responses that are evidence-based. The main aim of this project is to design a question answering system that retrieves relevant research papers, generates evidence-based responses, and validates reliability using biomedical language models.

1.2 Problem Definition

Due to the rapid growth of biomedical research literature, it is challenging to quickly find accurate information in large numbers of scientific papers. Existing search systems rely on keyword-based information retrieval. This makes it necessary to read many scientific papers to find the required information. This is time-consuming. Existing question-answering systems that rely on large language models do not necessarily provide answers that are fully supported by scientific evidence. The objective of the project is to design an effective Retrieval-Augmented Generation (RAG)-based biomedical question-answering system that not only provides answers but also checks the answers to ensure reliability and accuracy.

1.3 Scope and Motivation

The domain in which the system is used is related to biomedical literature, including journals, research papers, preprints, and abstracts. The main purpose behind developing this system is to help research professionals, educators, and clinicians access correct scientific information in a fast manner..

The main reason for developing this project is on the increasing rate of literature in the field of medicine. The increasing rate of literature is a time-consuming process, as different papers are required to be analyzed to retrieve information. The main aim of builting this project is to retrieve information with the help of automated tools by analyzing different scientific documents and on implementing the concept of multi-document summarization.

1.4 Objectives

The objectives of our project is:

- To develop a RAG system that provides factually grounded answers for Researchers by retrieving content from verified biomedical literature, eliminating hallucination in generated responses.
- To ensure transparency and trustworthiness in generated answer by providing citation and faithfulness evaluation.
- To improve relevancy of answer ensuring generated responses accurately address the user's query.

1.5 Assumptions

- The RAG system assumes that there are enough articles in the biomedical domain available in reliable open sources for the retrieval.

- It is assumed that questions are in user query format and belongs to biomedical domain.
- The system supposes the access to the database during the retrieval.
- It is supposed that the API calls are protected from attacks such as unauthorized access.

1.6 Societal / Industrial Relevance

This project greatly affects the productivity of biomedical research by providing validated and efficient ways of acquiring information in a timely manner. It also assists medical professionals, researchers, and data scientists in making evidence-based decisions. The industrial extension of this project may include drug discovery, clinical trials, and summarization of publications.

1.7 Organization of the Report

The report is organized into four chapters, each describing a specific part of the project and its development.

- **Chapter 1– Introduction:**
  This chapter presents an overview of the project, explaining the background, motivation, problem statement, objectives, and relevance of the proposed system. It builds the foundation for understanding the purpose and importance of developing a RAG-based biomedical research assistant.
- **Chapter 2– Literature Survey:**
  This chapter discusses about the literature regarding retrieval-augmented generation systems, biomedical information retrieval, and document validation approaches. This chapter aims to present essential findings of the related literature, compare the methodologies, and identify the gaps that the proposed system intends to fill.

- **Chapter 3– System Design:**

  This chapter aims to tell the design and architecture of the proposed system. It describes the modules, algorithms, and workflows involved, embedding with Nomic AI, and verification using ClinicalBERT. The hardware and software requirements, project timeline, and responsibilities of team members are also included.

- **Chapter 4- Results and Discussions**

  This chapter discusses the results achieved from the implementation of the proposed system for answering biomedical questions. It will show the system output for a query by a user and how relevant documents are retrieved from the biomedical field. The results will help evaluate the performance of the proposed system.

- **Chapter 5– Conclusion:**

  This chapter concludes the design phase by summarizing the outcomes achieved and highlighting the strengths of the proposed system. It also tells how the proposed system is useful for researchers, doctors aiming to provide the answers for the questions they are looking for.

1.8 Summary of the Chapter

This chapter presents a biomedical question-answering system that aims to assist users in efficiently retrieving relevant and accurate information from massive amounts of scientific literature.

To achieve this, the proposed system utilizes a retrieval-augmented generation (RAG) model that helps in retrieving relevant literature and answering questions with evidence and references. Additionally, it presents the aim of creating an effective and modular question-answering system that provides accurate and trustworthy answers with corresponding confidence scores. Overall, it demonstrates the significance of the proposed question-answering system and provides the basis for the subsequent chapters of the report.

Chapter 2

Literature Survey

This chapter discusses the existing research on biomedical question answering systems and retrieval-augmented generation techniques and their applications in the medical and clinical research domain. The literature survey focuses on recent advancements in large language model based question answering, retrieval pipelines using vector embeddings, and domain-specific biomedical models that enhance both answer accuracy and response reliability. The works reviewed provide insights into embedding architectures, document chunking strategies, retrieval mechanisms, and generation models that have been applied to biomedical text understanding and evidence-based answer generation.

2.1 Deep Learning Based Chatbot Architecture for Medical Diagnosis and Treatment Recommendation [1]

This paper addresses the problem of providing accessible medical advice and treatment recommendations through an automated conversational system. The authors propose a deep learning based chatbot that first classifies whether a user's input is medical or general in nature using a Naive Bayes classifier. Once the query is identified as medical, the system extracts symptoms from the user's natural language input and uses machine learning models including Decision Trees and Support Vector Classifiers to analyze the symptoms and predict possible diseases. The chatbot then recommends appropriate treatments based on the predicted condition. The system is designed to be lightweight and accessible, making it suitable for users who need quick preliminary health guidance without visiting a doctor. The paper demonstrates reasonable accuracy on a curated set of medical symptom-disease mappings and presents the chatbot as a practical tool for early-stage medical assistance in resource-limited environments.

2.1.1 Strengths

- Introduces an accessible chatbot interface that allows users to ask health related questions in natural language, making it user friendly and easy to interact with.
- Uses a combination of multiple machine learning classifiers which improves the accuracy of disease prediction by considering different aspects of the user's symptoms.

2.1.2 Weakness

- Relies entirely on static training data and pre-defined classification models, meaning the system cannot answer questions about new or recent medical findings.
- Does not retrieve from any external literature or research papers, so responses are limited to what the model was trained on and cannot be traced back to any published evidence.

2.1.3 Relevance to Project

This paper is relevant to our system as it establishes the foundation of using a conversational interface for handling biomedical queries in natural language. While this system relies on fixed classifiers, our project extends this idea by replacing the static prediction approach with a dynamic RAG pipeline that retrieves real published research papers and generates evidence-backed answers, addressing the core limitation of not being grounded in actual literature.

2.2 A Novel RAG Framework with Knowledge-Enhancement for Biomedical Question Answering [2]

This paper tackles the challenge of improving the reliability and accuracy of biomedical question answering systems using retrieval-augmented generation. The authors identify that standard RAG systems often struggle with biomedical text because the retrieved content may not always be semantically aligned with the specific intent of the query. To address this, they propose a RAG-Chain framework that enhances the retrieval process by incorporating domain-specific knowledge at multiple stages of the pipeline. The system uses sentence-level document chunking to break biomedical papers into smaller, more meaningful units and applies the BGE-M3 embedding model to generate dense vector representations of each chunk. These embeddings are then used to retrieve the most relevant content, which is combined with the user query and passed to a large language model to generate a coherent, context-supported answer. The RAG-Chain approach structures the retrieval and generation process as a sequential chain of operations, ensuring that each stage builds on the output of the previous one to progressively refine the quality of the final response.

2.2.1 Strengths:

- Introduces a knowledge-enhanced RAG pipeline that goes beyond basic retrieval by structuring the process into a chain, which improves the quality and coherence of the generated answers.
- Uses sentence-level chunking which preserves more granular context from the source documents compared to larger fixed-length chunks, resulting in more precise retrieval.

2.2.2 Weakness

- Does not include any reranking step after initial retrieval, meaning the top retrieved chunks may not always be the most relevant ones for the specific query being asked.
- Lacks a dedicated validation module to verify whether the generated answer is actually grounded in the retrieved documents, leaving no way to measure the trustworthiness of the response.

2.2.3 Relevance to Project

This paper is directly relevant to our system as it demonstrates the effectiveness of a structured RAG pipeline for biomedical question answering. Our system builds upon this approach by adding a multi-stage retrieval process that includes MMR-based diversity filtering and Cross-Encoder reranking on top of the initial FAISS retrieval, and further strengthens it with a ClinicalBERT validation step that this system lacks, ensuring every generated answer is verified for faithfulness and relevancy before being returned to the user.

2.3 Biomedical Literature Question Answering System Using Retrieval-Augmented Generation [3]

This paper presents one of the most architecturally similar systems to our system among the surveyed works. The authors propose a RAG-based biomedical question answering system that retrieves relevant information from a curated collection of PubMed articles and medical encyclopedias. The system uses the MiniLM sentence embedding model to convert both the user query and document chunks into dense vector representations, and then applies FAISS vector search to identify the most semantically similar chunks for a given query. These retrieved chunks are then passed to a fine-tuned Mistral-7B language model, which has been specifically adapted for biomedical text to generate accurate and contextually relevant answers. The authors evaluate the system on standard biomedical QA benchmarks and demonstrate that fine-tuning the generation model on domain-specific data leads to measurable improvements in factual accuracy compared to using a general-purpose language model. The paper presents this system as a practical and scalable solution for assisting researchers and clinicians in quickly finding answers to complex biomedical questions from large volumes of published literature.

2.3.1 Strengths

- Uses semantic embeddings combined with FAISS vector search which enables fast and accurate similarity-based retrieval from large collections of biomedical documents.
- Fine-tuning the Mistral-7B model on biomedical data significantly improves the factual accuracy and contextual relevance of the generated answers compared to using a general-purpose model.

2.3.2 Weakness

- Does not apply any diversity filtering or reranking on the retrieved chunks, which means the context passed to the model may contain redundant or less relevant information that could affect answer quality.
- The system does not compute any confidence or grounding scores, so users have

no way of knowing how well the generated answer is supported by the retrieved literature.

2.3.3 Relevance to Project

This paper shares the closest architectural similarity with our system in terms of combining FAISS-based retrieval with an LLM for answer generation. It validates the core retrieval and generation approach used in our system and highlights the importance of domain-adapted embeddings. Our project improves on this by replacing single-stage retrieval with a three-stage pipeline using FAISS, MMR, and Cross-Encoder reranking, and by adding ClinicalBERT-based validation to provide interpretable confidence scores that this system does not offer.

2.4 Developing a Virtual Diagnosis and Health Assistant Chatbot Leveraging LLaMA3 [4]

This paper introduces a virtual health assistant chatbot aimed at providing users with preliminary medical advice and health information through natural language interaction. The authors leverage the LLaMA3 large language model as the core generation engine and fine-tune it on the MedPalm medical dataset, which contains a large collection of medical question-answer pairs covering a wide range of clinical topics and diagnostic scenarios. This fine-tuning process significantly enhances the model's ability to understand medical terminology, interpret symptom descriptions, and generate clinically relevant responses. One of the notable contributions of this paper is the integration of multimodal support through the SigLIP Vision-Text Transformer, which allows the chatbot to process not only text-based queries but also medical images such as X-rays or skin lesion photographs alongside text descriptions. This makes the system considerably more versatile than text-only medical chatbots and opens up possibilities for visual symptom analysis. The paper evaluates the system on a diverse set of medical queries and demonstrates strong performance in terms of response coherence, clinical accuracy, and user satisfaction scores compared to baseline chatbot systems.

2.4.1 Strengths

- Fine-tuning on the MedPalm dataset gives the model a strong understanding of medical terminology and diagnostic patterns, enabling it to handle a wide range of health related queries effectively.
- The integration of multimodal support through the SigLIP Vision-Text Transformer allows the system to process both text and medical images, making it more versatile than text-only systems.

2.4.2 Weakness

- The system generates answers directly from the fine-tuned model without retrieving from any external or live literature, making it prone to hallucination especially for queries about recent medical developments not covered in training.
- Since the responses are entirely model-generated without retrieval, there are no references or citations provided with the answers, which reduces transparency and makes it difficult for users to verify the information.

2.4.3 Relevance to project

This paper is relevant to our system as it highlights the limitations of relying solely on a fine-tuned language model for medical question answering without any retrieval component. The hallucination problem and lack of citations identified in this system directly motivates the design choice in our project to strictly ground every answer in retrieved research papers from Europe PMC and to provide source references alongside every response, ensuring transparency and medical reliability that this system lacks.

2.5 RAG-BioQA: A Retrieval-Augmented Generation Framework for Long-Form Biomedical Question Answering [5]

This paper proposes a comprehensive RAG-based framework specifically designed to generate detailed, long-form answers to complex biomedical questions. The authors argue that existing biomedical QA systems tend to produce short, incomplete answers that do not fully address the depth and complexity of research-level medical queries. To overcome this, they develop a pipeline that uses BioBERT embeddings, which are pre-trained on a large corpus of biomedical text, to generate highly domain-specific vector representations of both queries and document chunks. These embeddings are indexed using FAISS for efficient vector-based retrieval, and the system searches across curated biomedical datasets including PubMedQA and MedQuAD to find the most relevant question-answer pairs for a given query. The retrieved content is then passed to a FLAN-T5 model that has been fine-tuned using LoRA, a parameter-efficient fine-tuning technique that adapts the model to the biomedical domain without the computational overhead of full fine-tuning. The system is evaluated on standard biomedical benchmarks and demonstrates strong performance in generating long-form, contextually rich answers while maintaining computational efficiency. The paper presents this system as a scalable and resource-conscious solution for biomedical research assistance.

2.5.1 Strengths

- Using BioBERT embeddings which are specifically pre-trained on biomedical text ensures that the semantic representations used for retrieval are domain-aware and better suited for biomedical queries than general-purpose embeddings.
- The use of LoRA fine-tuning on FLAN-T5 allows the system to achieve strong generation performance while keeping the computational cost low, making it more efficient than full fine-tuning approaches.

2.5.2 Weakness

- The system relies on pre-existing static QA datasets like PubMedQA and MedQuAD for retrieval rather than fetching live research papers, which means it cannot answer queries about findings published after the dataset was collected.
- Similar to other systems in this survey, it does not include any post-generation validation step to assess whether the generated answer is faithful to the retrieved content or relevant to the original user query.

2.5.3 Relevance to Project

This paper is the most closely related work to our system in terms of overall goals and architecture, as both systems use RAG with biomedical-specific embeddings and FAISS for retrieval. It serves as a strong baseline for comparison and reinforces the importance of domain-specific embeddings in biomedical retrieval. Our system advances beyond this by retrieving papers dynamically in real time from Europe PMC rather than relying on static datasets, and by incorporating a full post-generation validation pipeline using ClinicalBERT that computes faithfulness and relevancy scores which this system does not provide.

2.6 Research Gaps Identified

From the reviewed literature, the following gaps have been identified:

- Most existing biomedical question answering systems rely on pre-trained models or fine-tuning models without incorporating information from recent research literature, which makes it difficult for them to provide answers based on recent published results.
- None of the reviewed systems apply a multi-stage retrieval pipeline that combines initial vector-based search, diversity filtering through Maximal Marginal Relevance, and cross-encoder reranking, resulting in retrieved content that may be redundant or insufficiently relevant to the query.
- There is limited emphasis on transparency and citation in biomedical AI systems, with most systems generating answers without providing references to the source papers, making it difficult for researchers to verify or trace the information back to its origin.

These gaps justify the need for a real-time, retrieval-augmented biomedical research assistant that dynamically fetches live research literature, applies a multi-stage retrieval and reranking pipeline for accuracy and diversity, generates strictly evidence-grounded answers using a domain-aware language model, and validates every response with interpretable faithfulness and relevancy scores to ensure trustworthiness and transparency in biomedical research assistance.

2.7 Comparison of Reviewed Literature

Table 2.1: Comparison Table

Paper	**Focus**	**Methodology**	**Strengths**	**Limitations**
Deep Learning Based Chatbot Architecture for Medical Diagnosis and Treatment Recommendation	Medical diagnosis chatbot	Naive Bayes classifier, Decision Trees and SVC for disease prediction	Easy to use chatbot interface, multi-classifier approach improves prediction	No retrieval, limited to static training data, no evidence-backed responses
A Novel RAG Framework with Knowledge-Enhancement for Biomedical Question Answering	Biomedical QA using RAG	RAG-Chain with BGE-M3 embeddings and sentence-level chunking	Knowledge-enhanced pipeline, fine-grained sentence-level chunking	No reranking after retrieval, no answer validation or confidence scoring
Biomedical Literature Question Answering System Using Retrieval-Augmented Generation	Biomedical literature QA	MiniLM + FAISS retrieval + fine-tuned Mistral-7B generation	Fast semantic retrieval, domain-adapted generation model	No diversity filtering or reranking, no grounding or confidence scores
Developing a Virtual Diagnosis and Health Assistant Chatbot Leveraging LLaMA3	Virtual health assistant chatbot	LLaMA3 fine-tuned on MedPalm + SigLIP multimodal support	Strong medical understanding, supports text and image inputs	No retrieval from literature, hallucination-prone, no citations in output
RAG-BioQA: A Retrieval-Augmented Generation Framework for Long-Form Biomedical Question Answering	Long-form biomedical QA	BioBERT + FAISS + LoRA fine-tuned FLAN-T5 generation	Domain-specific embeddings, computationally efficient with LoRA	Static QA datasets only, no real-time retrieval, no post-generation validation

The table shows a comparative analysis of the existing biomedical question-answering systems in terms of focus, methodology, strength, and weakness. From the table, it is clear that though there are advanced models used in the existing approaches, there is still a lack of integration with updated sources, proper retrieval, as well as the validation of the answers, which are important in the real world.

2.8 Summary of Chapter

The literature review highlights the current state of biomedical question answering systems, focusing on retrieval-augmented generation pipelines, domain-specific language models such as BioBERT and ClinicalBERT, and large language model based answer generation approaches using models like Mistral-7B, FLAN-T5, and LLaMA3. While these foundational works demonstrate strong performance in retrieving relevant biomedical content and generating contextually accurate responses, they reveal significant gaps in their application to real-time, evidence-grounded research assistance. The reviewed systems lack a focus on multi-stage retrieval pipelines that ensure both relevance and diversity, and validation mechanisms that verify the faithfulness and reliability of generated answers, and the ability to dynamically fetch live research literature. These gaps emphasize the need for a tailored biomedical research assistant that bridges high-accuracy retrieval with a multi-stage reranking pipeline, a grounded generation module, and an interpretable validation framework. The insights gained from this review directly inform the design choices for the RAG pipeline, the ClinicalBERT-based validation module, and the overall system architecture presented in the next chapter.

Chapter 3

System Design

This chapter explains how BioResearch AI,Biomedical Research Assistant is designed. The main goal of this system is to help users find accurate answers to biomedical questions by searching through real research papers. The system fetches actual published literature, breaks it down into chunks, and uses it to generate reliable, evidence-backed answers. This chapter explains the working of the system by using system architecture, sequence diagram and the key modules of the system.

3.1 Overall Architecture

The overall system architecture of the RAG-Based Biomedical Research Assistant combines natural language processing with multi-document retrieval, and large language models, to generate accurate responses to biomedical research queries. This system retrieves scientific literature in real time and provides evidence-supported answers that are properly referenced. The overall flow is as follows:

3.1.1 Query Input and Processing:

The system first receives a user query in the structured form: *"What is the [research keyword] of [intervention/subject] in [disease/condition] from papers published in [year/timeframe]?"*. Regular expressions are applied to process the query and extract key biomedical terms such as the intervention, condition, and timeframe. These extracted keywords are used to identify the most relevant research papers.

3.1.2 Document Retrieval:

The extracted keywords are used to construct API calls to the Europe PMC [6] database. Based on the specified timeframe and biomedical terms, the system retrieves relevant research papers. Metadata including title, authors, journal name, and publication year are collected, and the corresponding PDF documents are downloaded locally for further processing.

3.1.3 Document Parsing and Embedding:

The downloaded PDFs are parsed using the PyMuPDF [7] library. Informative sections such as Introduction, Methods, Results, and Discussion are retained, while non-informative sections like Abstract, References, and Acknowledgements are removed. The cleaned text is divided into fixed-length overlapping chunks to preserve contextual continuity. Each chunk is associated with metadata such as paper title and section name, and then converted into 768-dimensional vector embeddings using the Nomic AI embedding model (*nomic-embed-text-v1*) [8].

3.1.4 Vector Storage:

The generated embeddings, along with their metadata such as source paper name, section type, and chunk length, are stored in a FAISS [9] vector index. This structure enables efficient similarity search and comparison of document chunks during retrieval.

3.1.5 Semantic Search and Chunk Retrieval:

The user query is also converted into an embedding using the Nomic AI model. The FAISS index is searched using cosine similarity to retrieve the most relevant chunks. These retrieved chunks are further re-ranked using Maximal Marginal Relevance (MMR) to ensure diversity. A Cross-Encoder reranker is then applied to improve the accuracy of scoring each chunk with respect to the query.

3.1.6 Answer Generation:

The selected chunks are further processed and scored using a PubMedBERT [10] based bi-encoder followed by a cross-encoder reranking mechanism. The most relevant and diverse sentences are selected through MMR to construct the final context. This context is then provided to the Qwen2 [11] large language model along with a prompt, enabling the model to generate a final answer based solely on the retrieved biomedical evidence without relying on external knowledge.

3.1.7 Answer Validation and Output:

The generated answer is validated using ClinicalBERT by computing the cosine similarity between the generated response and the retrieved source chunks. Faithfulness and answer relevancy scores are calculated to ensure that the generated response is accurate and grounded in the retrieved biomedical literature. Finally, the system outputs the generated answer along with references to the corresponding source papers.

3.2 Design Diagrams

3.2.1 Architecture Diagram

The architecture shows a representation of the end-to-end workflow , which includes both retrieval and generation. This process begins with the query, which is processed using keyword extraction. This is then followed by the retrieval of documents from a temporary database. A detailed processing of these documents is then done, which includes metadata extraction, such as title, authors, and keywords, as well as semantic chunking, where the documents are broken down into meaningful chunks, and vector embeddings.The resulting embedding is then stored in a FAISS vector database, which facilitates the search for similar vectors. The user query is converted into an embedding, allowing the system to retrieve the most relevant chunks based on semantic similarity. These retrieved chunks are then fed into the answer generation and validation module, where a language model is used to create an accurate response. The output of the system is not only an accurate response but also includes references, making it reliable.

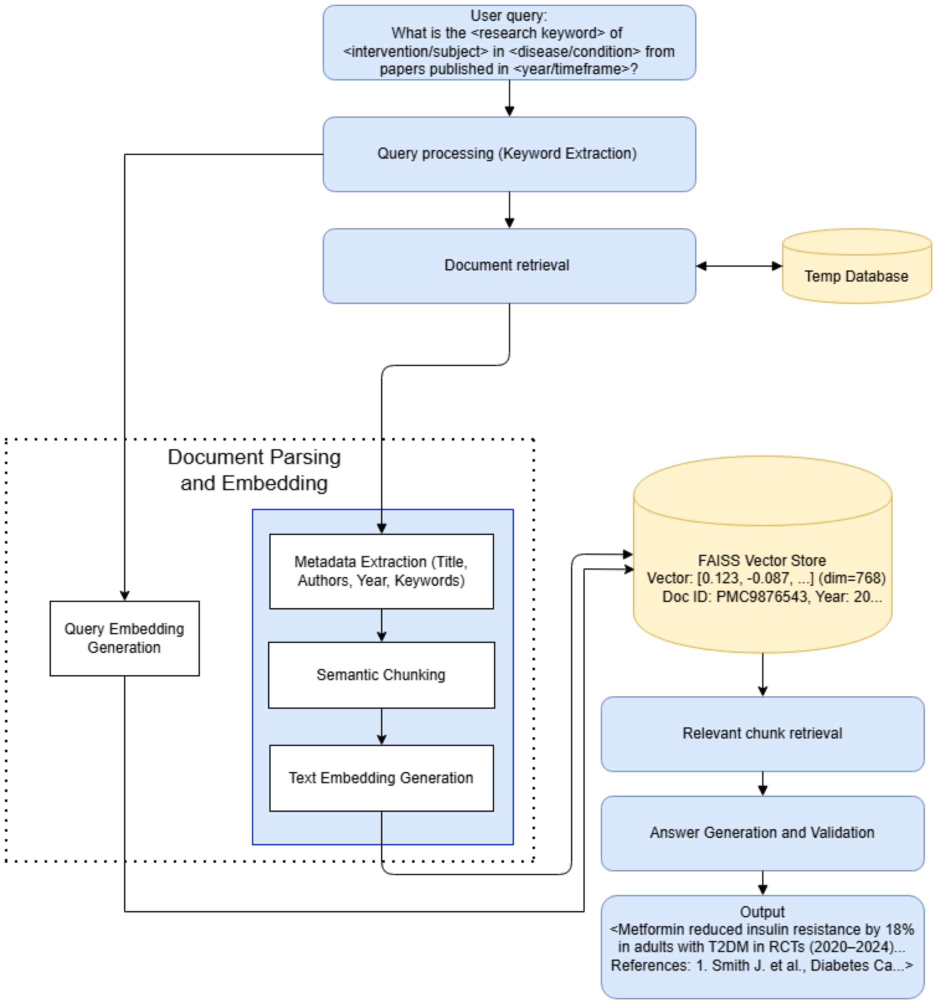


Figure 3.1: System Architecture of the Biomedical QA System

3.2.2 Sequence Diagram

The sequence diagram shows the step-by-step interaction among the system's components in the processing of the user's query. It also shows the flow of communication among the modules and how each component of the system is working together for the smooth execution of the task and the production of the final result. It also explains the coordination among the different stages of the system.

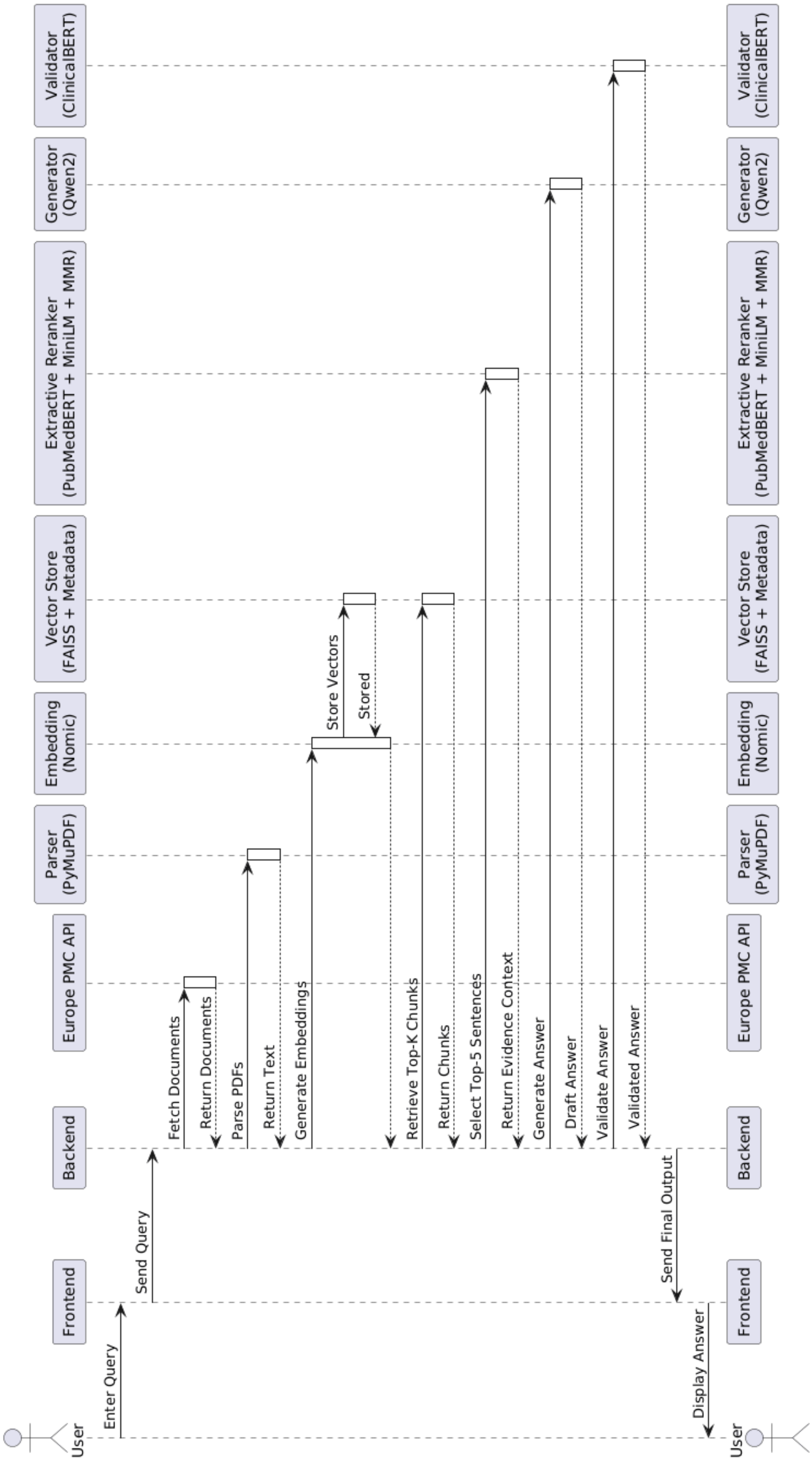


Figure 3.2: Sequence Diagram of the Biomedical QA System

3.2.3 System Workflow

The sequence diagram shows the system workflow where the user interacts with the frontend, the backend, and the processing modules that are required to produce the correct biomedical research answers. For instance, a user types a biomedical query on the chat interface with the following heading:
"What is the [research keyword] of [intervention] in [condition] from papers published in [year range]?" This query is received by the frontend and submitted to the backend using FASTAPI. The backend initially performs the query parsing and the regular expressions are applied to the query, to retrieve the required keywords like intervention, condition and timeframe. These terms are sent to the Europe PMC database through REST API to pull relevant papers. The relevant PDFs are downloaded and they are locally stored. PDF files are converted to text by parsing using PyMuPDF. The cleaned text is chunked into overlapping fixed-length chunks . These chunks are then passed to an embedding module which uses the Nomic AI embedding model to embed each chunk into dense vector representations. The vectors and their metadata are passed to the FAISS vector database and stored there.The query is also embedded using the same Nomic AI pipeline.We search the FAISS index for chunks that are most similar to those retrieved from the query. The retrieved top chunks are passed to an Extractive reranker. Within this, we extract and score sentences, using a PubMedBERT-based bi-encoder, and then reranking it using a MiniLM [12] cross-encoder and MMR and top 5 sentences are passed along with a structured prompt to Qwen2 language model for answer generation. The model outputs a final answer based on biomedical content that was retrieved. Once the answer is produced it is passed to ClinicalBERT along with user query and retrieved chunks to validate the answer . The system calculates the faithfulness score and answer relevancy score and is printed in the backend. The validated answer is passed to the frontend. The final answer is then shown on the chat interface to the user as well as a link to the respective source papers. This is the basic workflow of the application which ensure accurate source backed answers to biomedical queries.

3.3 Module Division

The BioResearch AI system consists of four main modules, each responsible for a specific part of the pipeline. Taken together, these modules provide a complete end-to-end workflow starting with the user request to provide a validated biomedical answer.

3.3.1 Module 1: Query Processing and Document Retrieval.

The Query Processing and Document Retrieval module is the first module of the system.When a user inputs a query with the form: "What is the [research keyword] of [intervention] in [condition] from papers published in [year/timeframe]?"It is processed using regular expressions to obtain 4 main parameters: the research keyword, intervention, medical condition, and publication period. If a given year range is given , we split it into start and end year; if a single year is given, both are set to that year; and if no year is mentioned, the year range is set to last five years. These selected parameters are then used as keywords which are sent to the Europe PMC database through their REST API, for the retrieval of relevant and freely available papers. If the Europe PMC search does not return or shows no results, the system automatically transitions with a PubMed fallback to remain reliable. When matching papers are found, the system extracts links for each PDF download, downloads the PDF files locally, and saves the abstract with a metadata file (title, authors, journal, and publication year) to facilitate further processing in the next module.

3.3.2 Module 2: Document Parsing and Embedding.

The document parsing and embedding module converts the PDF research papers that are downloaded into a vector format for the system. After downloading PDF, it is first processed to structured markdown text by using the PyMuPDF library. Then the text is checked line by line for headers for specific sections, and then a section head type is mapped to known section types such as Introduction, Methods, Results, Discussion, and Conclusion by regular expression pattern matching. Sections that are not useful for finding, such as abstract, references, bibliography, and acknowledgements are found and removed from further processing. The rest of the sections are cleaned to remove noise, i.e. DOI links, citation numbers, page numbers and formatting will be cleared out. The cleaned content is then segmented into overlapping fixed-length, 1500 character, overlapping 200 character fixed-length chunks, which can keep the contextual meaning between contiguous chunks within and beyond the cut-offs. Each chunk is kept along with its metadata such as the referential name of the paper, section type, chunk ID, and length. Once all the chunks are ready, they are fed to the Nomic AI embedding model (nomic-embed-text-v1) for converting each chunk to a 768-dimensional dense vector structure. The obtained embeddings and corresponding metadata are stored in JSON format and are passed for the next step of semantic search and retrieval.

3.3.3 Module 3: Semantic Search and Relevant Chunk Retrieval

Semantic Search and Relevant Chunk Retrieval is the module that retrieves relevant chunks from the stored embeddings.This module loads loads all the chunk embeddings from the JSON file generated by the previous module and builds a FAISS index using inner product similarity after L2 normalization.The query is transformed into a vector, using the same Nomic AI embedding model. This query vector is used for searching the FAISS index and selecting the top 30 most similar chunks as initial candidates by cosine similarity. To prevent returning redundant or repetitive results, Maximal Marginal Relevance (MMR) is then executed with a lambda value of 0.5, balancing relevance and diversity by penalizing chunks that too resemble ones already sampled, resulting in the top 10 most relevant and diverse chunks. These 10 chunks proceed to a final reranking step with a Cross-Encoder model (ms-marco-MiniLM-L-6-v2), which ranks each combination of query-chunk pairs in a more accurate manner than the original vector search. The chunks are then sorted by their reranking scores and the top 5 highest scoring chunks are returned with their rank, score, source paper name, section type, and text content, so that they can be passed on to the answer generation module.

3.3.4 Module 4: Answer Generation and Validation

The Answer Generation and Validation module is the final module of the system that generates a reliable, evidence-based answer based on the segments retrieved chunks.The top ranked chunks generated by the previous module are passed to this module and the chunks are broken down into individual sentences using NLTK's sentence tokenizer, removing duplicates and any sentences that are too short to be meaningful. Each sentence is then scored for relevance to the user query using a PubMedBERT-based bi-encoder model which produces normalized embeddings for the query and each sentence and calculates their cosine similarity. The top 25 most relevant sentences are passed through a Cross-Encoder model , which evaluates each query-sentence pair together in a more precise manner, generating refined relevance scores. MMR is then applied with a lambda value of 0.7 to select the final top 5 sentences, balancing between relevance and diversity to avoid redundancy in the context. These selected sentences are combined in their original source order to form a clean context string, which is then formatted into a structured prompt. The system prompt tells the Qwen2 language model, which is running locally via Ollama, to answer strictly using the provided context without relying on any outside knowledge, while the user template presents the context and question in a clear format.After generating the answer, it is validated with ClinicalBERT[13] , in which the answer is assessed against the top 5 retrieved source chunks using cosine similarity calculations of their mean-pooled embeddings. The average of these similarity scores is used for the faithfulness score, and if it meets or exceeds 0.7, the answer is considered grounded.Answer relevancy score is also calculated which is the semantic similarity between the original user query and the generated answer.The final validated answer is presented to the user along with the references.

3.4 Tools and Technologies

3.4.1 Software Requirements

Programming Languages

- Python: It is used for implementing backend logic, model integration, and API handling.
- React: It is used for building the user interface and managing frontend interactions.

Frameworks and Libraries

- FastAPI: Backend framework that coordinates the entire pipeline from receiving user queries to delivering validated biomedical answers.
- PyMuPDF: It is used for parsing and extracting text from PDF documents.

- FAISS: It is used for storing vector embeddings and performing similarity search.

APIs

- Europe PMC: External API used to retrieve relevant biomedical literature and research papers based on the user query.

AI/ML Models

- Nomic AI: It is used for chunk and query embedding.
- PubMedBERT: A bi-encoder model trained on the PubMed dataset used for extracting top sentences related to the query.
- MiniLM: A cross-encoder model used for reranking the top retrieved sentences.
- Qwen2: A large language model used for natural language generation of the final biomedical answer.
- ClinicalBERT: A BERT model trained on clinical datasets used for answer validation.

3.4.2 Hardware Requirements

- 8 GB RAM minimum is required for running embedding models, FAISS indexing, and the Qwen2 LLM.
- Stable Internet Connection is essential for real-time API access and model interactions.
- Optional GPU for faster processing and improved model inference performance.

3.5 Dataset Identified

- Biomedical Research Articles: Real-time retrieval of peer-reviewed articles and abstracts from Europe PMC, covering various biomedical domains relevant to user queries.
- Metadata: Extracted metadata for each article, including title, authors, publication year, keywords, DOI, and PMC ID.

3.6 Work breakdown and Responsibilities

The RAG-Based Biomedical Research Assistant system is divided into three main modules, each assigned to different team members as follows:

- **Query Processing and Document Retrieval**
  Team Member: Abhay
  Responsibilities:
  - Accept structured biomedical queries from the user.
  - Extract keywords and biomedical entities using regular expressions.
  - Retrieve relevant research articles from EuropePMC using keyword-based search.
  - Filter and store the top 5 relevant articles for downstream processing.
- **Document Parsing, Chunking, Embedding, and Vector Storage**
  Team Member: Abel
  Responsibilities:
  - Parse full-text PDFs and abstracts using PyMuPDF.
  - Split documents into chunks for manageable embedding.
  - Generate embeddings for each chunk using Nomic AI embedding.
  - Store embeddings and metadata in the FAISS vector database for similarity-based retrieval.
  - Embed the user query and retrieve top-k relevant document chunks from FAISS.
- **Answer Generation and Verification**
  Team Members: Achyuth and Albin
  Responsibilities:
  - Retrieve the top 5 sentences from the shortlisted chunks and prepare context for Qwen.
  - Generate coherent answers using Qwen2 LLM by giving a structured prompt along with the retrieved context.

- Verify answers against retrieved chunks and the user query to ensure source-backed responses.
- Print the generated answer as output with confidence scores and references.

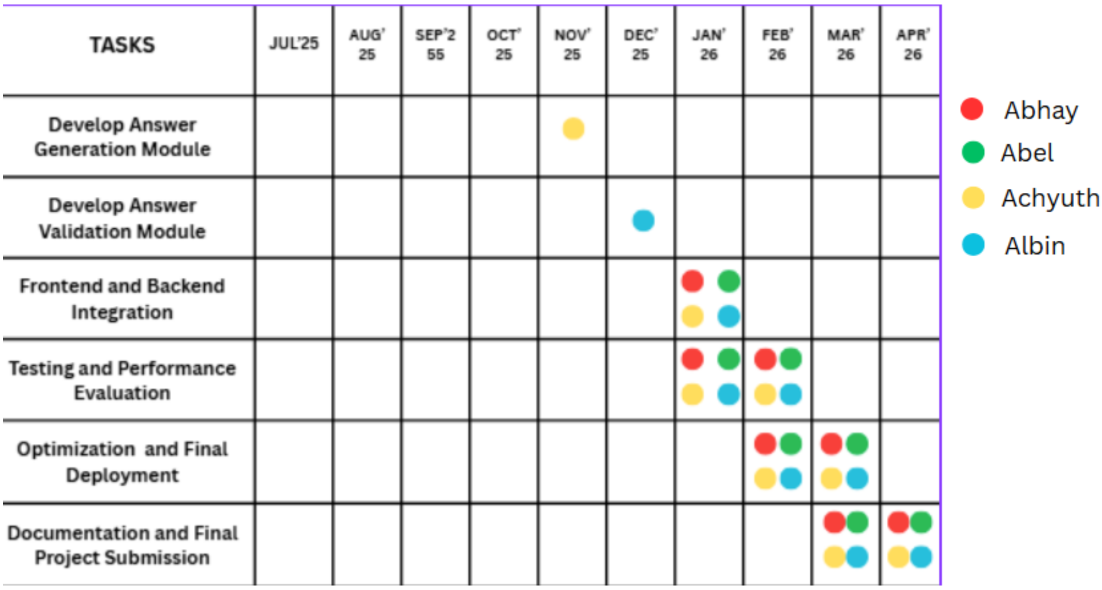
Module	Team Members	Timeline
Query Processing and Document Retrieval	Abhay	Oct–Nov
Document Parsing and Embedding	Abel	Oct–Nov
Answer Generation	Achyuth	Nov-Dec
Answer Validation	Albin	Nov-Dec
Frontend Implementation	All	January
Integration	All	January
System Testing	All	February

Table 3.1: Module-wise Work Division and Timeline

3.7 Key Deliverables - Expected Output

- The system generates a concise, evidence-based answer to the user's biomedical query using the Qwen2 language model, which produces the response strictly from the retrieved research literature without relying on any external knowledge.
- The Supporting Evidence section lists the citations and references of the documents,journal articles, or data sources. It includes details like title, URL,DOI of the research papers that were used to generate the final answer.This ensures reliability of the system's output.
- The faithfulness score is computed by ClinicalBERT by measuring the cosine similarity between the generated answer and the top 3 retrieved source chunks, indicating how well the answer is grounded in the actual retrieved biomedical literature.
- The relevancy score is also computed by ClinicalBERT by measuring the cosine similarity between the original user query and the generated answer, indicating how directly and accurately the answer addresses what the user actually asked.

3.8 Project Timeline



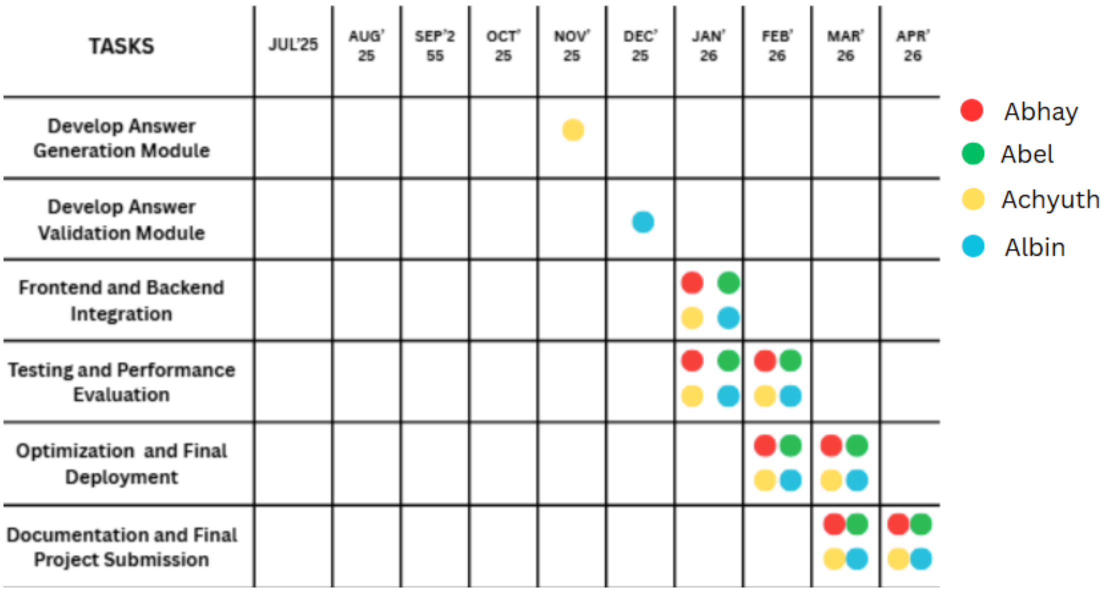


Figure 3.3: Gantt Chart

3.9 Summary of chapter

This chapter presents the system design and implementation plan of the project. It describes the system architecture and system workflow of the system and shows design diagrams of the system to indicate the structure and functionality of the system, like architecture diagram and sequence diagram. Furthermore, the chapter presents the 4 major modules of the system namely query processing and document retrieval, document parsing and embedding, semantic search and chunk retrieval, and answer generation and validation. It discusses the tools and technologies used, work distribution among team members, key deliverables, and the project timeline

Chapter 4

Results and Discussions

The results presented in this chapter demonstrate the performance of the proposed biomedical question answering system. The system was evaluated using multiple biomedical queries to analyze how effectively it retrieves relevant documents and generates accurate responses.

The evaluation focuses on three key aspects: retrieval quality, answer generation, and overall system reliability. Each stage of the pipeline contributes to the final output, and therefore, the performance is analyzed in a step-by-step manner.

4.1 Results and Functions

4.1.1 Query Processing and Document Retrieval

The query processing stage plays a critical role in understanding user intent. The system extracts key biomedical terms from the query and transforms them into a structured representation. This ensures that the retrieval process focuses on the most relevant concepts.

The document retrieval module uses embedding-based similarity search to identify relevant documents from the dataset. The quality of retrieved documents directly affects the final answer. If irrelevant documents are retrieved, the generated response may also become inaccurate.

The system shows consistent performance in retrieving contextually relevant biomedical documents, which improves the grounding of the generated answers.

4.1.2 Chunk Embedding and FAISS Construction

The retrieved documents are divided into smaller chunks to improve processing efficiency. Chunking helps the system focus on fine-grained information rather than processing entire documents at once.

Each chunk is converted into a vector representation using embedding models. These vectors capture the semantic meaning of the text. The FAISS library is used to store and index these vectors, enabling fast similarity search.

Table 4.1: Comparison of Text Embedding Models

Model	Parameters	Max Seq Length	MTEB	LoCo	Jina LC	Open Weights	Local
nomic-embed-text-v1	137M	8192	62.39	85.53	54.16	Yes	Yes
text-embedding-3-small	N/A	8192	62.26	82.40	58.21	No	No
text-embedding-3-large	N/A	8192	64.59	79.40	58.69	No	No
E5-Mistral-7B-Instruct	7B	4096	**66.60**	**87.80**	N/A	Yes	Limited
BGE-large-en-v1.5	335M	512	64.23	83.12	57.00	Yes	Yes
Cohere embed-v3	N/A	512–4096	63.47	81.32	56.80	No	No

Table 4.1 shows that E5-Mistral-7B-Instruct achieves the highest performance scores. However, it has higher computational requirements. On the other hand, nomic-embed-text-v1 provides a balanced trade-off between performance and efficiency, making it suitable for real-world deployment.

4.1.3 Answer Generation and Validation

After retrieving the most relevant chunks, the system generates answers using a language model. The generated output is then validated against the retrieved context to ensure factual correctness.

Table 4.2: LLM as Judge Evaluation Results Across LLM Models

Model	Faithfulness	Answer Rel.	Completeness	Fluency	Overall
Qwen2	1.00	0.90	0.94	1.00	**0.960**
Mistral	1.00	0.90	0.92	0.98	0.950
Llama 3	1.00	0.86	0.88	1.00	0.935
FLAN-T5 Large	1.00	0.75	0.50	0.58	0.745

Table 4.2 shows comparison of different LLM models using Google Gemini as the judge. Qwen2 achieves the highest overall score of 0.960. It performs consistently across all metrics, especially in fluency and faithfulness. Mistral and Llama 3 also perform well, while FLAN-T5 shows lower performance in completeness and fluency.

These results indicate that Qwen2 is the most suitable model for answer generation in this system.

4.2 Discussions

The proposed system integrates retrieval and generation to provide accurate answers to biomedical queries. It reduces the need for manual search through research articles and improves accessibility to biomedical knowledge.

Table 4.3: Comparison of RAG-based Biomedical QA Systems

System	**Faithfulness**	**Context Rel.**	**Answer Rel.**
Proposed System	**0.97**	0.83	0.78
MedRAG	0.93	**0.86**	0.82
Garg et al.	0.90	0.85	0.78
RAG-Chain	0.92	0.80	**0.84**
RAG-BioQA	0.89	0.84	0.77
Self-RAG	0.91	0.82	0.80
FLARE	0.90	0.83	0.79

The evaluation results show that the proposed system achieves the highest faithfulness score of 0.97. This indicates that the generated answers are strongly grounded in the retrieved biomedical context.

The system achieves a context relevance score of 0.83, which is comparable to other methods. MedRAG achieves the highest score of 0.86 due to its extensive retrieval strategy.

For answer relevance, the proposed system achieves 0.78, while RAG-Chain achieves 0.84. This difference is mainly due to the reasoning-based design of RAG-Chain.

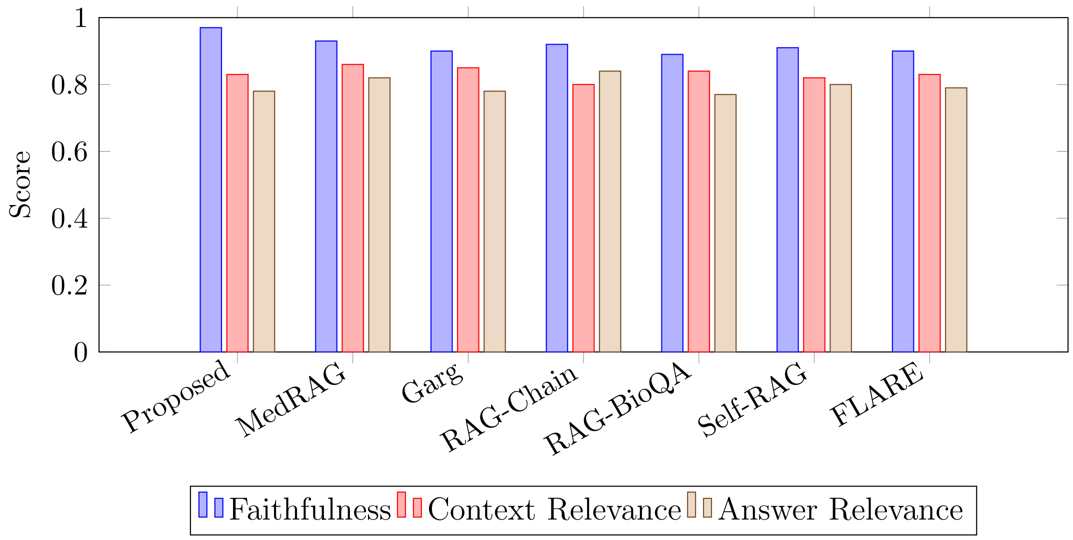


Figure 4.1: Performance comparison of biomedical RAG systems across RAGAS metrics

Figure 4.1 provides a visual comparison of the performance of different RAG-based systems across key evaluation metrics. The proposed system clearly achieves the highest faithfulness score of 0.97, indicating strong grounding of generated answers in the retrieved biomedical context.

MedRAG shows strong performance in context relevance with a score of 0.86, while RAG-Chain achieves the highest answer relevance of 0.84 due to its reasoning-based approach. Other models such as Self-RAG and FLARE show balanced performance across all metrics.

The graph highlights that the proposed system maintains a consistent balance across all evaluation metrics while outperforming other systems in faithfulness. This makes it more reliable for biomedical applications where accuracy and factual correctness are critical.

Overall, the proposed system maintains a strong balance between accuracy and efficiency. It prioritizes factual correctness and minimizes hallucinations, which is critical for biomedical applications.

The results confirm that the system is reliable and suitable for real-world biomedical question answering tasks.

Chapter 5

Conclusions & Future Scope

5.1 Conclusion

The project proves that Large Language Models, when integrated with Retrieval Augmented Generation, are effective in answering queries through information retrieval from biomedical literature. The system helps users retrieve accurate information by processing relevant documents obtained from biomedical literature, thus helping users understand biomedical information better.

The system helps users access biomedical information easily by providing users with information obtained directly from biomedical literature without having to go through numerous documents of biomedical research papers. The project proves that integrating document retrieval systems with language models is effective in helping users access accurate biomedical information through various applications, including question answering systems, retrieval systems, etc., which can be of immense help to users, including researchers, students, and practitioners in the biomedical field.

5.2 Future Scope

Currently, the system is focused on a particular domain. However, the system can be extended to accommodate other domains such as engineering, law, finance, scientific research, etc. This will make the system more versatile to a wider audience of users.

The system can be improved further by incorporating more sophisticated language models, more efficient document retrieval methods, a more interactive interface, the capability to update documents in real-time, etc.

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List of Publications

1. Saintgits College Of Engineering,"A Rag Based Biomedical Research Assistant" submitted to International Conference on Smart Communication and Sustainable Technologies (ICSCST 2026), under the guidance of Ms. Seema Safar

Appendix A: Presentation

BioResearch AI

A RAG based Biomedical Research Assistant

Project Guide: Ms Seema Safar

Team Members

1) Abel T John (U2203007)
2) Abhay P(U2203008)
3) Achyuth Pradeep (U2203015)
4) Albin Sam Alex(U2203025)

Contents

Problem Definition

Developing a biomedical research assistant that integrates Retrieval-Augmented Generation (RAG) with long-context Large Language Models (LLMs) to enable efficient exploration of large volumes of biomedical literature, providing fast, accurate, and context-aware answers to domain-specific queries while ensuring responses are grounded in relevant research papers with supporting evidence to reduce hallucination and improve reliability.

Problem Objective

- To develop a modular and interpretable RAG pipeline separating document retrieval, chunk-level search, and answer generation for improved reliability and scalability.
- Provide evidence-supported answers with references from biomedical research papers to improve transparency and trustworthiness.
- To provide interpretable confidence scores that quantify grounding and relevance to check whether the answer addresses the query.
- Use a lightweight and efficient generation module capable of producing accurate responses while maintaining computational efficiency and strong performance compared to similar models.

Purpose and Need

- Researchers often need to analyze large volumes of biomedical research articles, which can be time-consuming, especially in critical domains such as healthcare and clinical research.
- Large Language Models can generate answers quickly, but these responses are not always grounded in reliable sources, creating the need for mechanisms that verify and support generated information.
- Providing confidence scores and supporting references helps users quickly understand how strongly an answer is supported by retrieved documents.
- This project improves trust and reliability in AI-assisted research tools by ensuring that responses are relevant, evidence-based, and suitable for critical decision-making environments.

Literature Survey

Deep Learning Based Chatbot Architecture for Medical Diagnosis and Treatment Recommendation [1]
Conference: International Conference on Advanced Computing Technologies and Applications (ICACTA) (2023)
Authors: G. Rajani et al.

• Uses a deep learning based chatbot to provide medical advice and treatment recommendations.

• Classifies user queries using a Naïve Bayes classifier to determine whether the input is medical or general.

• Uses machine learning models such as Decision Trees and Support Vector Classifiers to analyze symptoms and predict possible diseases.

Literature Survey

• Implemented a chatbot-based interface to allow users to ask biomedical queries in natural language.

• Uses natural language query processing to understand user questions and generate responses automatically.

• Extends the idea by using Retrieval-Augmented Generation (RAG) to retrieve biomedical research literature and generate context-supported answers.

Literature Survey

A Novel RAG Framework with Knowledge-Enhancement for Biomedical Question Answering [2]
Conference: IEEE International Conference on Bioinformatics and Biomedicine (BIBM) (2024)
Authors: Y. Du, Z. Wang et al.

- Proposes a RAG-Chain framework to improve biomedical question answering using large language models.
- Uses sentence-level document chunking and embedding models (BGE-M3) to represent biomedical documents for retrieval.
- Combines retrieved context with a Large Language Model to generate context-supported answers.

Literature Survey

Paper Title: Biomedical Literature Question Answering System Using Retrieval-Augmented Generation (RAG) [3]
Source: arXiv Research Paper (2025)
Authors: M. Garg et al.

• Proposes a RAG-based biomedical question answering system that retrieves information from biomedical literature such as PubMed articles and medical encyclopedias.

• Uses semantic embeddings (MiniLM) and FAISS vector search to retrieve the most relevant biomedical document chunks for a query.

• Generates responses using a fine-tuned Mistral-7B language model, improving factual accuracy and contextual relevance.

Literature Survey

Paper Title: Developing a Virtual Diagnosis and Health Assistant Chatbot Leveraging LLaMA3 [4]
Conference: International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS) (2024)
Authors: A. Patel et al.

- Develops a virtual health assistant chatbot using the LLaMA3 large language model to provide preliminary medical advice and health information.
- The model is fine-tuned using the MedPalm medical dataset to improve understanding of medical queries and diagnostic responses.
- Integrates the SigLIP Vision-Text Transformer to support multimodal inputs such as medical images along with text queries.

Literature Survey

• Implemented a chatbot-based system to handle biomedical queries and generate automated responses.

• Uses large language models to understand user queries and generate answers.

• Extends this approach by integrating Retrieval-Augmented Generation (RAG) to retrieve biomedical literature before generating responses.

Literature Survey

Paper Title: RAG-BioQA: A Retrieval-Augmented Generation Framework for Long-Form Biomedical Question Answering [5]
Source: arXiv Research Paper (2026)
Authors: L. Y. Panchumarthi et al.

- Proposes a RAG-based framework for generating long-form biomedical answers using retrieved biomedical knowledge.
- Uses BioBERT embeddings and FAISS vector indexing to retrieve relevant biomedical question–answer pairs from datasets such as PubMedQA and MedQuAD.
- Generates responses using a LoRA fine-tuned FLAN-T5 model, enabling efficient biomedical answer generation.

Literature Survey

• Implemented a Retrieval-Augmented Generation pipeline to answer biomedical queries using research literature.

• Uses embedding-based retrieval and semantic chunking to identify relevant biomedical information.

• Uses Europe PMC research articles as the dataset to retrieve biomedical literature and generate context-supported answers.

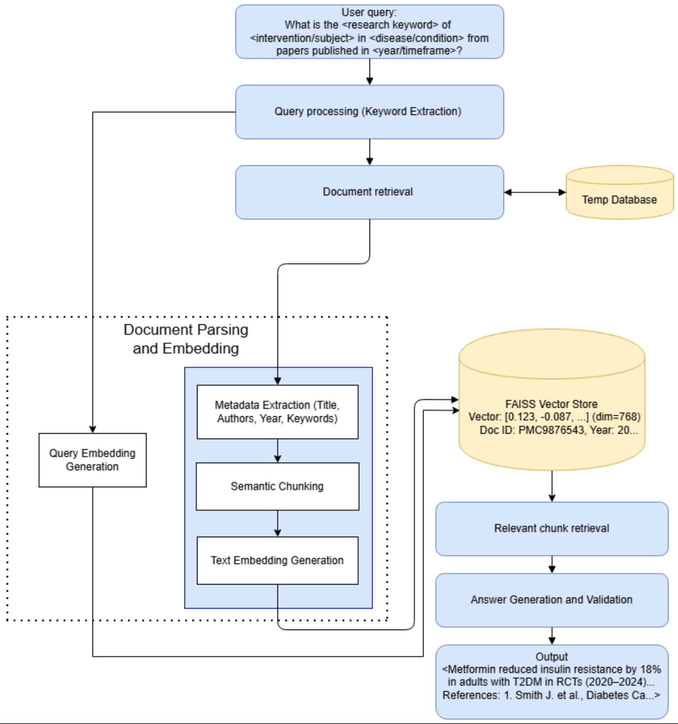
Proposed Method

- The proposed system is a Retrieval-Augmented Generation (RAG) based framework designed to assist users in understanding biomedical research literature and answering medical queries.

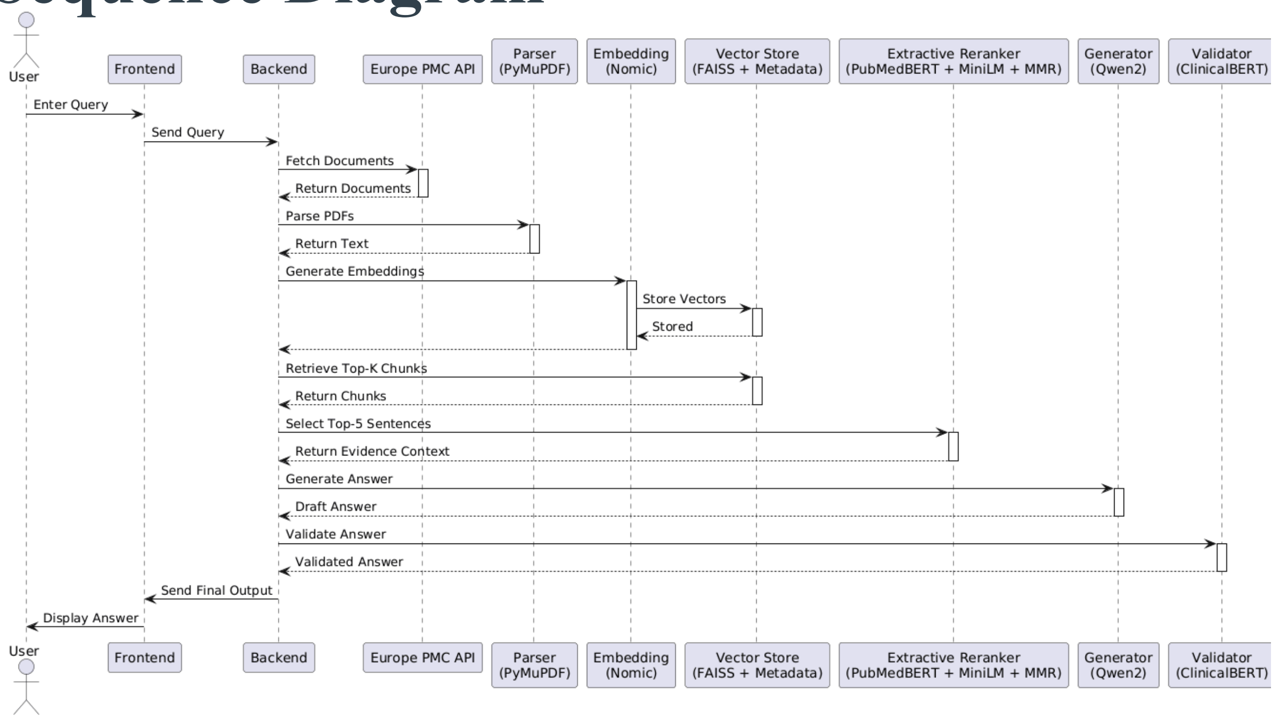
- The system first analyzes the user query and retrieves relevant biomedical research papers using keyword-based document selection. The retrieved documents are then divided into smaller meaningful sections through semantic chunking, and the most relevant chunks are identified using embedding-based retrieval.

- The retrieved context is then used by a Large Language Model to generate responses that are supported by biomedical research information. By combining document selection, chunk-level retrieval, and response generation within a single framework, the system helps produce clear and context-supported biomedical answers.

Architecture Diagram



Sequence Diagram



Module Description

1. Query processing and Document retrieval
2. Document Parsing and Embedding
3. Semantic Similarity Search and Relevant Chunk Retrieval
4. Answer generation and Validation

Query Processing And Document Retrieval

- **Query Processing**

Converts natural language queries into structured parameters using regular expressions.

Intended query structure:

"What is the <research keyword> of <intervention> in <condition> from papers published in <year/time range>?"

Example: "What is the effect of aspirin in heart disease from papers published in 2018–2022?"

Parameter Extraction :

Extracts key biomedical terms from the query for document search.

These parameters guide the retrieval system to identify relevant research papers.

Example: intervention → aspirin, condition → heart disease.

Timeframe Identification :

Supports both single-year queries and year ranges.

Example: "published in 2015–2020" → start year 2015, end year 2020.

Applies a default timeframe if no year is specified (papers from current year − 5 to current year)

- **Document Retrieval**

The system retrieves biomedical research papers from the Europe PMC database using the processed query parameters.

The query is combined with publication year filters and open-access constraints to ensure relevant and accessible research articles.

The system uses the Europe PMC REST API to fetch metadata such as title, authors, journal, and publication year.

After retrieving search results, the system downloads the PDF files and stores them locally for further processing.

Document Parsing and Embedding

- **Document Parsing and Cleaning**

The system converts downloaded PDF research papers into structured markdown text using the pymupdf4llm library.

The parser identifies section headers such as Introduction, Methods, Results, Discussion, and Conclusion using pattern matching.

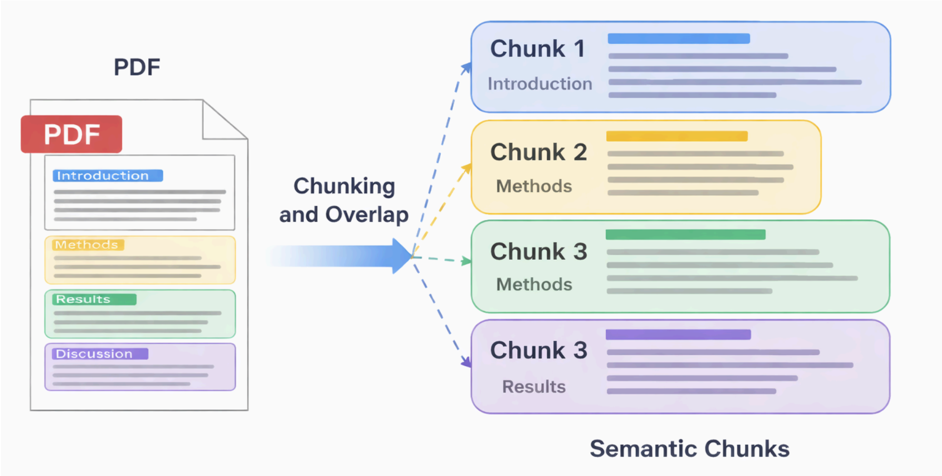
Sections that do not contribute useful information for retrieval, such as Abstract, References, and Acknowledgements, are excluded from further processing.

Extracted text is cleaned to remove noise such as DOI links, citation numbers, page numbers, and unnecessary formatting artifacts.

- **Section-Aware Overlapping Fixed-Length Chunking**

The cleaned content is divided into overlapping text chunks to preserve contextual continuity during semantic retrieval.

Each chunk is stored with metadata including source paper name, section type, and chunk length for later embedding and similarity search.



- **Text Embedding Generation**

Each text chunk is first combined with contextual metadata such as the paper title and section name to provide meaningful context before embedding.

Example:

Prepared text sent for embedding:

Paper: Aspirin Therapy Study |

Section: Methods |

Content: Patients were administered aspirin daily for...

The prepared text is converted into dense vector embeddings using the Nomic embedding model (nomic-embed-text-v1).

Text chunk:

"Patients were administered aspirin daily to study its effect on platelet aggregation."

Embedding vector representation:

[0.124, -0.087, 0.562, ...]

Semantic Similarity Search and Relevant Chunk Retrieval

- **Semantic Similarity Search**

  A FAISS vector index is used to efficiently compare the query embeddings with the stored chunk embeddings.

  The system retrieves the top candidate chunks based on cosine similarity, identifying the most semantically relevant content.

- **Relevant Chunk Retrieval**

  The retrieved candidate chunks are refined using Maximal Marginal Relevance (MMR) to remove redundant results and ensure diversity in the selected chunks.

  A Cross-Encoder reranker (ms-marco-MiniLM-L-6-v2) is then applied to evaluate the relevance between the query and each chunk more accurately.

  The system finally selects the top-ranked chunks, which are passed to the answer generation module.

Answer generation and Validation

- **Context Extraction and Selection**

  The retrieved document chunks are first divided into individual sentences using sentence tokenization.

  Each sentence is evaluated for relevance using a bi-encoder model (PubMedBERT based SentenceTransformer) to measure semantic similarity with the query.

  A cross-encoder reranker is then applied to further refine relevance scores by evaluating the query and sentence together.

  Maximal Marginal Relevance (MMR) is used to select the most relevant and diverse sentences, which are combined to form the final context for answer generation.

- **Answer Generation**

  The selected context and query are formatted into a structured prompt using predefined system and user templates.

  The prompt is passed to the Qwen2 large language model, which generates the final answer based strictly on the retrieved context.

  The model is instructed to avoid external knowledge and rely only on the provided biomedical evidence, ensuring grounded and reliable responses.

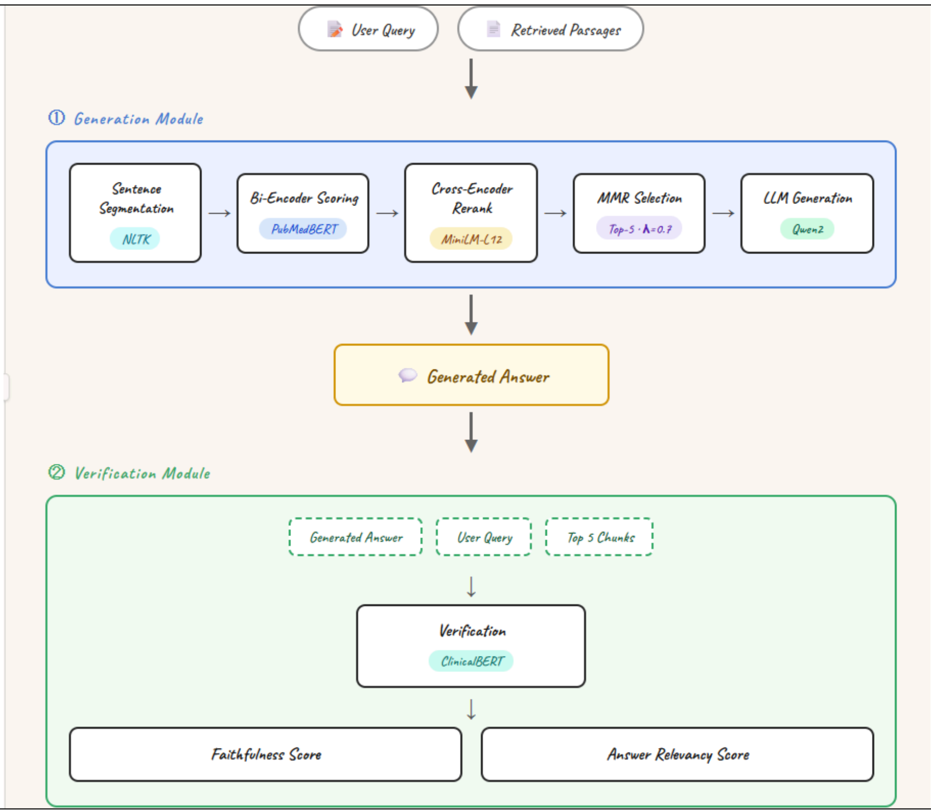
- **Answer Validation**

  The generated answer is verified using Bio_ClinicalBERT, a domain-specific biomedical language model.

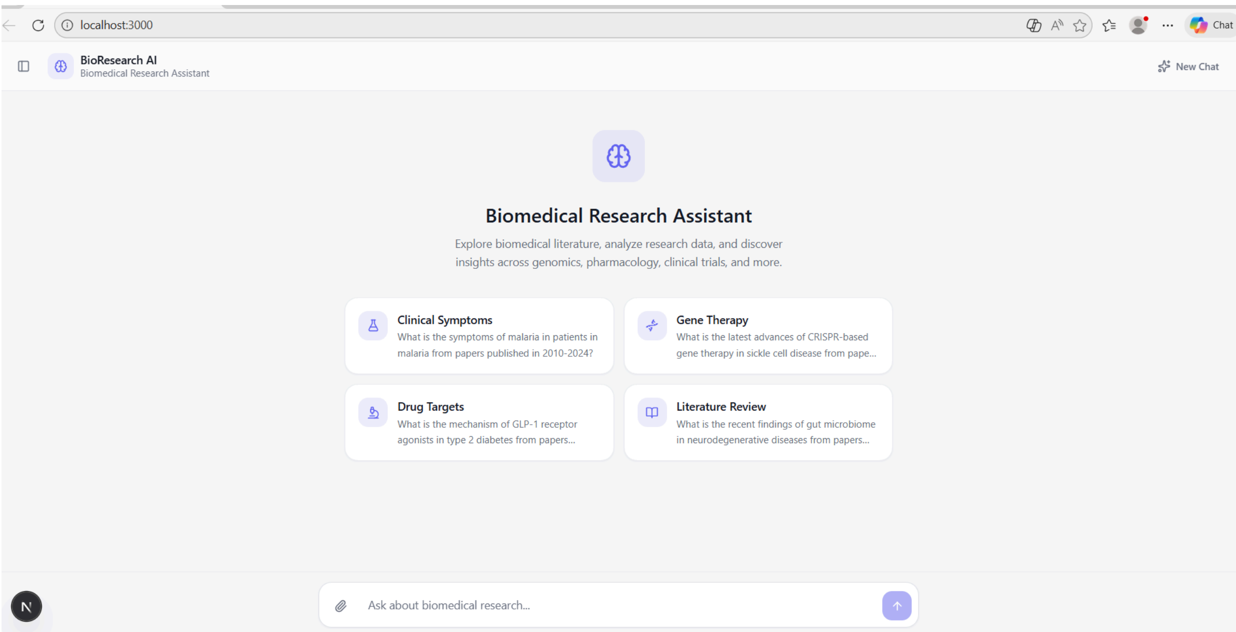
  The system computes cosine similarity between the generated answer and the retrieved source chunks to measure how well the answer is grounded in the evidence.

  The similarity scores from the top retrieved chunks are averaged to determine whether the answer is medically grounded or requires additional evidence.

  The system also evaluates the semantic similarity between the user query and the generated answer using ClinicalBERT embeddings.


User Query
Retrieved Passages
① Generation Module
Sentence Segmentation
NLTK
Bi-Encoder Scoring
PubMedBERT
Cross-Encoder Rerank
MiniLM-L12
MMR Selection
Top-5 · λ=0.7
LLM Generation
Qwen2
Generated Answer
② Verification Module
Generated Answer
User Query
Top 5 Chunks
Verification
ClinicalBERT
Faithfulness Score
Answer Relevancy Score

Results


localhost:3000
Chat
BioResearch AI
Biomedical Research Assistant
New Chat
Biomedical Research Assistant
Explore biomedical literature, analyze research data, and discover insights across genomics, pharmacology, clinical trials, and more.
Clinical Symptoms
What is the symptoms of malaria in patients in malaria from papers published in 2010-2024?
Gene Therapy
What is the latest advances of CRISPR-based gene therapy in sickle cell disease from pape...
Drug Targets
What is the mechanism of GLP-1 receptor agonists in type 2 diabetes from papers...
Literature Review
What is the recent findings of gut microbiome in neurodegenerative diseases from papers...
Ask about biomedical research...

Query Parsing and Document retrieval

```
================================================================================
🚀 FULL PIPELINE TRIGGERED
================================================================================

================================================================================
STEP 1: Query Parsing
================================================================================
Europe PMC query : recent findings gut microbiome neurodegenerative diseases
Timeframe        : 2018-2025

================================================================================
STEP 2: Document Retrieval (Europe PMC)
================================================================================
Downloading papers: 100%|██████████████████████████████| 3/3 [02:30<00:00, 50.12s/it]
Retrieval stats: {'retrieved': 3, 'pdf_downloaded': 3, 'abstracts_saved': 3}

================================================================================
STEP 3: PDF Cleaning, Section Extraction & Chunking
================================================================================

 Processing: From_Broad_Spectrum_Health_to_Targeted_Prevention_.pdf
    Found 14 total sections
    Keeping:   General – 'General'
    Keeping:   General – 'Xinyun Zhang** **** **, Qinghua Zeng** *'
    Excluding: ABSTRACT – 'Abstract'
    Keeping:   General – '1. Introduction'
    Keeping:   General – '2. Literature Search Methodology'
    Keeping:   General – '3. The Role of Functional Foods in Chron'
    Keeping:   General – 'Bioactive** **Main Mechanisms** **Primar'
    Keeping:   General – 'Core Component** **Traditional Rice Endo'
    Keeping:   General – 'Scientific Advancement'
    Keeping:   General – 'Compound** **Meta-Analyses** **Strength'
    Keeping:   General – '5. Conclusions'
    Keeping:   General – '6. Perspectives and Outlook'
    Keeping:   General – 'Hypothesis** **Mechanistic Focus** **Mod'
    Excluding: REFERENCES – 'References'
    Keeping 12, excluding 2
    Saved 82 chunks → papers\From_Broad_Spectrum_Health_to_Targeted_Prevention_.json
```

```
Processing: More_than_Dysbiosis__Imbalance_in_Humoral_and_Neur.pdf
   Found 13 total sections
   Keeping:   General – 'General'
   Keeping:   General – 'and Almira Kustubayeva** **[1,2,]** ***'
   Excluding: ABSTRACT – 'Abstract'
   Keeping:   General – '1. Introduction'
   Keeping:   General – '2. Multimodal Bidirectional Gut-Brain In'
   Keeping:   General – 'Gut-Brain Axis'
   Keeping:   General – 'Gut-Brain Axis'
   Keeping:   General – '3. Unbalanced Gut-Brain Axis in AD'
   Keeping:   General – '4. Perspectives for AD Therapy Through t'
   Keeping:   General – '5. Discussion'
   Keeping:   General – '6. Perspective and Future Directions'
   Keeping:   General – '7. Summary'
   Excluding: REFERENCES – 'References'
   Keeping 11, excluding 2
   Saved 48 chunks → papers\More_than_Dysbiosis__Imbalance_in_Humoral_and_Neur.json

Processing: Potential_application_of_mono__dual__and_triple_.pdf
   Found 14 total sections
   Keeping:   General – 'General'
   Keeping:   General – 'Potential application of mono-, dual-, a'
   Keeping:   General – '1 Introduction'
   Keeping:   General – '2 GLP-1 RA cardiovascular protection and'
   Keeping:   General – '3 Comparison of different GLP-1 RA drugs'
   Keeping:   General – '4 Adverse effects of GLP-1 RA in DFU pat'
   Keeping:   General – '5 Personalized treatment recommendations'
   Keeping:   General – '6 Conclusion and clinical implications'
   Keeping:   General – 'Author contributions'
   Excluding: REFERENCES – 'References'
   Excluding: ACKNOWLEDGEMENTS – 'Funding'
   Keeping:   General – 'Conflict of interest'
   Keeping:   General – 'Generative AI statement'
   Keeping:   General – 'Publisher's note'
   Keeping 12, excluding 2
   Saved 35 chunks → papers\Potential_application_of_mono__dual__and_triple_.json

Processing: Psychobiotics_at_the_Frontiers_of_Neurodegenerativ.pdf
   Found 12 total sections
   Keeping:   General – 'General'
   Keeping:   General – 'and María Elena Sánchez-Pardo** **[1,]**'
```

```
  Keeping:   General – 'General'
  Keeping:   General – 'and María Elena Sánchez-Pardo** **[1,]**'
  Excluding: ABSTRACT – 'Abstract'
  Keeping:   General – '1. Introduction'
  Keeping:   General – 'Disorders'
  Keeping:   General – 'Pathways'
  Keeping:   General – '4. Psychobiotics: Definition and Key Mic'
  Keeping:   General – 'Condition** **Probiotic Strain(s)** **St'
  Keeping:   General – 'Condition** **Probiotic Strain(s)** **St'
  Keeping:   General – '5. Future Perspectives in Psychobiotics '
  Keeping:   General – '6. Conclusions'
  Excluding: REFERENCES – 'References'
  Keeping 10, excluding 2
  Saved 46 chunks → papers\Psychobiotics_at_the_Frontiers_of_Neurodegenerativ.json

Processing: Research_progress_in_precision_medicine_for_type_2.pdf
  Found 14 total sections
  Keeping:   General – 'General'
  Keeping:   General – 'Research progress in precision medicine '
  Keeping:   General – 'T2DM diagnosis and treatment trends'
  Keeping:   General – 'Overview of GLP-1'
  Keeping:   General – 'Biology of GLP-1-secreting l cells'
  Keeping:   General – 'GLP-1 related drug development process'
  Keeping:   General – 'Advances in metabolic surgery'
  Excluding: ABSTRACT – 'Summary and future directions'
  Keeping:   General – 'Author contributions'
  Excluding: ACKNOWLEDGEMENTS – 'Funding'
  Excluding: REFERENCES – 'References'
  Keeping:   General – 'Conflict of interest'
  Keeping:   General – 'Generative AI statement'
  Keeping:   General – 'Publisher's note'
  Keeping 11, excluding 3
  Saved 36 chunks → papers\Research_progress_in_precision_medicine_for_type_2.json

Processing: Synergistic_therapeutic_strategies_for_metabolic_d.pdf
  Found 15 total sections
  Keeping:   General – 'General'
  Keeping:   General – 'Synergistic therapeutic strategies for m'
  Keeping:   General – '1 Introduction'
  Keeping:   General – '2 Shared genetic predisposition of MASH '
```

```
  Keeping:   General – '3 Shared pathophysiology of MASH and T2D'
  Keeping:   General – '4 Preclinical insights into the mechanis'
  Keeping:   General – '5 Clinical therapeutic approaches in MAS'
  Keeping:   General – '6 Conclusion and future perspectives'
  Keeping:   General – 'Author contributions'
  Excluding: ACKNOWLEDGEMENTS – 'Funding'
  Excluding: REFERENCES – 'References'
  Excluding: ACKNOWLEDGEMENTS – 'Acknowledgments'
  Keeping:   General – 'Conflict of interest'
  Keeping:   General – 'Generative AI statement'
  Keeping:   General – 'Publisher's note'
  Keeping 12, excluding 3
  Saved 73 chunks → papers\Synergistic_therapeutic_strategies_for_metabolic_d.json

Processing: Targeting_gut_brain_immune_axis_in_amyotrophic_lat.pdf
  Found 16 total sections
  Keeping:   General – 'General'
  Keeping:   General – 'Targeting gut-brain-immune axis in amyot'
  Keeping:   General – '1 Introduction'
  Keeping:   General – '2 Hypothesis'
  Keeping:   General – '3 Supporting evidence'
  Keeping:   General – '4 Discussion'
  Keeping:   General – '5 Immune dysregulation and neuroinflamma'
  Keeping:   General – '6 Conclusion'
  Keeping:   General – 'Data availability statement'
  Keeping:   General – 'Author contributions'
  Excluding: REFERENCES – 'References'
  Excluding: ACKNOWLEDGEMENTS – 'Funding'
  Excluding: ACKNOWLEDGEMENTS – 'Acknowledgments'
  Keeping:   General – 'Conflict of interest'
  Keeping:   General – 'Generative AI statement'
  Keeping:   General – 'Publisher's note'
  Keeping 13, excluding 3
  Saved 36 chunks → papers\Targeting_gut_brain_immune_axis_in_amyotrophic_lat.json

Processing: The_application_of_fecal_microbiota_transplantatio.pdf
  Found 13 total sections
  Keeping:   General – 'General'
  Keeping:   General – 'The application of fecal microbiota tran'
  Keeping:   General – '1 Introduction'
  Keeping:   General – '2 The relationship between the MGBA and '
```

```
   Keeping:   General — 'General'
   Keeping:   General — 'The application of fecal microbiota tran'
   Keeping:   General — '1 Introduction'
   Keeping:   General — '2 The relationship between the MGBA and '
   Keeping:   General — '3 Application of FMT and oral probiotic '
   Keeping:   General — '4 Conclusions and future perspectives'
   Keeping:   General — 'Author contributions'
   Excluding: ACKNOWLEDGEMENTS — 'Funding'
   Excluding: ACKNOWLEDGEMENTS — 'Acknowledgments'
   Keeping:   General — 'Conflict of interest'
   Keeping:   General — 'Generative AI statement'
   Excluding: REFERENCES — 'References'
   Keeping:   General — 'Publisher's note'
   Keeping 10, excluding 3
   Saved 112 chunks → papers\The_application_of_fecal_microbiota_transplantatio.json

Processing: The_gut_microbiota_influences_neurodegenerative_di.pdf
   Found 13 total sections
   Keeping:   General — 'General'
   Keeping:   General — 'The gut microbiota influences neurodegen'
   Keeping:   General — '1 Introduction'
   Keeping:   General — '2 Gut microbiota-gut-brain axis'
   Keeping:   General — '3 The role of the gut microbiota in neur'
   Keeping:   General — '4 Therapeutic measures based on gut micr'
   Keeping:   General — '5 Discussion and conclusion'
   Keeping:   General — 'Author contributions'
   Excluding: REFERENCES — 'References'
   Excluding: ACKNOWLEDGEMENTS — 'Funding'
   Keeping:   General — 'Conflict of interest'
   Keeping:   General — 'Generative AI statement'
   Keeping:   General — 'Publisher's note'
   Keeping 11, excluding 2
   Saved 83 chunks → papers\The_gut_microbiota_influences_neurodegenerative_di.json
✅ Cleaned & chunked documents: 9
```

Chunk Embedding

```
STEP 4: Chunk Embedding
================================================================================

📄 Processing From_Broad_Spectrum_Health_to_Targeted_Prevention_.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (32 texts)...
   Embedding batch 3 (18 texts)...
   82 chunks embedded from From_Broad_Spectrum_Health_to_Targeted_Prevention_.json

📄 Processing More_than_Dysbiosis__Imbalance_in_Humoral_and_Neur.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (16 texts)...
   48 chunks embedded from More_than_Dysbiosis__Imbalance_in_Humoral_and_Neur.json

📄 Processing Potential_application_of_mono__dual__and_triple_.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (3 texts)...
   35 chunks embedded from Potential_application_of_mono__dual__and_triple_.json

📄 Processing Psychobiotics_at_the_Frontiers_of_Neurodegenerativ.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (14 texts)...
   46 chunks embedded from Psychobiotics_at_the_Frontiers_of_Neurodegenerativ.json

📄 Processing Research_progress_in_precision_medicine_for_type_2.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (4 texts)...
   36 chunks embedded from Research_progress_in_precision_medicine_for_type_2.json

📄 Processing Synergistic_therapeutic_strategies_for_metabolic_d.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (32 texts)...
   Embedding batch 3 (9 texts)...
   73 chunks embedded from Synergistic_therapeutic_strategies_for_metabolic_d.json

📄 Processing Targeting_gut_brain_immune_axis_in_amyotrophic_lat.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (4 texts)...
   36 chunks embedded from Targeting_gut_brain_immune_axis_in_amyotrophic_lat.json
```

```
📄 Processing Targeting_gut_brain_immune_axis_in_amyotrophic_lat.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (4 texts)...
   36 chunks embedded from Targeting_gut_brain_immune_axis_in_amyotrophic_lat.json

📄 Processing The_application_of_fecal_microbiota_transplantatio.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (32 texts)...
   Embedding batch 3 (32 texts)...
   Embedding batch 4 (16 texts)...
   112 chunks embedded from The_application_of_fecal_microbiota_transplantatio.json

📄 Processing The_gut_microbiota_influences_neurodegenerative_di.json
   Embedding batch 1 (32 texts)...
   Embedding batch 2 (32 texts)...
   Embedding batch 3 (19 texts)...
   83 chunks embedded from The_gut_microbiota_influences_neurodegenerative_di.json

🎉 Done — 551 total chunks embedded → chunk_embeddings.json
✅ Embeddings ready

================================================================================
STEP 5: FAISS Index Construction
================================================================================

================================================================================
STEP 6: Semantic Retrieval
```

```
mantic Retrieval
================================================================
00%|██████████████████████████████████████████████| 1/1 [00:01<00:00,  1.84s/it]

ore    | Section         | Source
--------------------------------------------------------------
259    | General         | The_application_of_fecal_microbiota_transplantatio.pdf
116    | General         | The_gut_microbiota_influences_neurodegenerative_di.pdf
090    | General         | The_gut_microbiota_influences_neurodegenerative_di.pdf
529    | General         | The_gut_microbiota_influences_neurodegenerative_di.pdf
098    | General         | The_gut_microbiota_influences_neurodegenerative_di.pdf
1, 'score': 5.259146213531494, 'text': '2.1 Gut microbiota involvement in PD\npathogenesis\n\nPD is a neurodegenerative disorder predominantly affecting\nmiddle-aged and elderly populations, clinically cha
dinal motor manifestations including resting tremor, rigidity,\n\nbradykinesia, and postural instability. Substantial evidence indicates\nthat non-motor symptoms, particularly gastrointestinal disturbances
ders, sleep abnormalities, sensory deficits, cognitive\nimpairment, and autonomic dysfunction—frequently precede motor\nmanifestations by several years and demonstrate progressive\nexacerbation with diseas
ng these non-motor manifestations, gastrointestinal\ndysfunction represents the most prevalent symptom complex in PD\npatients. The human gastrointestinal tract harbors a diverse microbial\necosystem that
dulates both motility and endocrine\nfunctions. This intestinal microbiota comprises a dynamic consortium\nof microorganisms, including bacteria, archaea, viruses, and\neukaryotic organisms (e.g., fungi an
ccupying\ndistinct ecological niches throughout the gastrointestinal lumen,\ncollectively termed the gut microbiome (Arumugam et al., 2011). The\nmetabolic activities and complex interactions of gut microb
y influence host physiological homeostasis and disease\nsusceptibility. As the “second genome” of the human body, the gut\nmicrobiome plays a fundamental role in both the initiation and\nprogression of neu
orders', 'source': 'The_application_of_fecal_microbiota_transplantatio.pdf', 'section': 'General'}, {'rank': 2, 'score': 5.116055011749268, 'text': 'Neurodegenerative diseases (NDDs) represent a complex cl
characterized by progressive loss of neuronal function or structure, including\nAlzheimer’s disease (AD), Parkinson’s disease (PD), and amyotrophic lateral sclerosis\n(ALS) (1). A common feature of these d
 misfolding and aggregation,\nleading to neuronal death and dysfunction, which in turn causes impairment of cognitive,\nmotor, or autonomic functions (2). In recent years, with the acceleration of global a
ce of NDDs has significantly increased, becoming a majo[illegible]ic\nburden. Although traditional research has made some progress in the pathological\nmechanisms and therapeutic strategies of NDDs
iology and\nmultifactorial interactions remain incomple[illegible]e pathogenesis of NDDs\ninvolves multiple pathways, including pathological protein aggregation, synaptic\n\n[Frontiers in Immunolog
ntiersin.org/journals/immunology) 01 [frontiersin.org](https://www.frontiersin.org)\n\nYang et al. [10.3389/fimmu.2025.1739329](https://doi.org/10.3389/fimmu.2025.1739329)\n\ndysfunction, protein homeostas
chondrial\ndysfunction, neuroinflammation, ferroptosis, and interactions\nbetween genetic and environmental factors (3–5). With advances\nin biology, the role of gut microbiota in NDDs has garnered\nwidesp
search indicates that gut microbiota\ninfluence brain function through multiple pathways, including\nneural, e', 'source': 'The_gut_microbiota_influences_neurodegenerative_di.pdf', 'section': 'General'}, {
: 4.089582920074463, 'text': 'Jiaheng Yang , Xiangzhun Song , Su Yan , Qianxun Li \n\nand Wenying Yang [*]\n\n1Clinical College of Medicine, Changchun University of Chinese Medicine, Changchun, China,\n\n2
roenterology, Jilin Province People’s Hospital, Changchun, Jilin, China\n\nThe pathogenesis of neurodegenerative diseases (NDDs), including Alzheimer’s\ndisease (AD), Parkinson’s disease (PD), and amyotrop
sis (ALS), is\n\ncomplex and multifactorial. Recent studies indicate that the microbiota-gut\nbrain axis (MGBA) plays a crucial role in the development and progression of\n\nNDDs. The MGBA concept reveals
ional regulatory network\n\nbetween the gut microbiota and the central nervous system (CNS), linking them\n\nthrough immune, neural, endocrine, and metabolic pathways. This review\n\nsummarizes the compone
ommunication pathways between\n\ngut microbiota and the brain, and mechanisms by which gut microbiota\ninfluence the onset and progression of NDDs. Finally, preclinical therapeutic\n\napproaches for NDDs a
uating preclinical trial data for\n\nprobiotics, prebiotics, and fecal microbiota transplantation.\n\nKEYWORDS\n\nAD, fecal microbiota transplantation, gut-microbiota-brain axis, NDDs, PD', 'source': 'The_
luences_neurodegenerative_di.pdf', 'section': 'General'}, {'rank': 4, 'score': 3.5293569564819336, 'text': 'es—that\ndirectly or indirectly shape neuronal plasticity and synaptic\nfunction (29). Research i
gut microbiota\ninfluences serotonin synthesis by regulating the tryptophan\nmetabolic pathway, thereby modulating mood and cognitive\nfunction (30).\nAdditionally, the gut microbiota regulates neuroinflam
europrotective mechanisms through vagus nerve\nactivation and immune modulation. For example, in AD models,\naltered gut microbiota reduced neuroinflammation and cognitive\nimpairment by increasing indole-
ls (31). The\nmicrobiota also contribute to neurodegenerative pathology by\ndynamically regulating BBB permeability, thereby controlling the\nCNS exposure to circulating metabolites and neurotransmitters (
position of the gut microbiota changes-a\ncondition known as dysbiosis-it may trigger a series of\npathophysiological responses (33). This process promotes\nsystemic immune activation (21, 34). Inflammator
n relayed to the CNS via immune and neuroendocrine routes,\nultimately driving neuroinflammation and neurodegeneration (33).\nAccumulating evidence underscores a strong association between\ngut microbiome
pathogenesis of several\n\nneurodegenerative disorders, including AD, PD, and multiple\nsclerosis (MS) (35).\n\n2.3 Core communication pathways\n\nMGBA constitutes a complex bidirectional communication\nne
ates information exchange between gut\nmicrobiota and the CNS through multipl', 'source': 'The_gut_microbiota_influences_neurodegenerative_di.pdf', 'section': 'General'}, {'rank': 5, 'score': 2.09761142730
teral Sclerosis [Internet]. (2025).\n[Available from: https://clinicaltrials.gov/study/NCT07017946.](https://clinicaltrials.gov/study/NCT07017946)\n\n177. Sproten R, Nohr D, Guseva D. Nutritional strategie
ut\nmicrobiome as a preventative and therapeutic approach in normal and pathological\nage-related cognitive decline: A systematic review of preclinical and clinical findings.\n[Nutr Neurosci. (2024) 27:104
/1028415x.2023.2296727](https://doi.org/10.1080/1028415x.2023.2296727)\n\n178. Gates EJ, Bernath AK, Klegeris A. Modifying the diet and gut microbiota to\nprevent and manage neurodegenerative diseases. Rev
33:767-87.\n[doi: 10.1515/revneuro-2021-0146](https://doi.org/10.1515/revneuro-2021-0146)\n\n179. Settanni CR, Ianiro G, Bibbo S, Cammarota G, Gasbarrini A. Gut microbiota\nalteration and modulation in psy
: current evidence on fecal\nmicrobiota transplantation. Prog Neuropsychopharmacol Biol Psychiatry. (2021)\n[109:110258. doi: 10.1016/j.pnpbp.2021.110258](https://doi.org/10.1016/j.pnpbp.2021.110258)\n\n[F
logy](https://www.frontiersin.org/journals/immunology) 17 [frontiersin.org](https://www.frontiersin.org)', 'source': 'The_gut_microbiota_influences_neurodegenerative_di.pdf', 'section': 'General'}]
```

Answer Generation and Validation

```
STEP 7: Hybrid Extractive + Abstractive Generation
================================================================================
INFO | Loading bi-encoder: pritamdeka/S-PubMedBert-MS-MARCO
INFO | Load pretrained SentenceTransformer: pritamdeka/S-PubMedBert-MS-MARCO
INFO | Loading cross-encoder: cross-encoder/ms-marco-MiniLM-L-12-v2
INFO | Cross-encoder loaded successfully.

 Extracted Evidence Sentences:
------------------------------------------------------------
[Rank 1] Accumulating evidence underscores a strong association between
gut microbiome dysbiosis and the pathogenesis of several
[Rank 4] Recent studies indicate that the microbiota-gut
brain axis (MGBA) plays a crucial role in the development and progression of

NDDs.
[Rank 5] Gates EJ, Bernath AK, Klegeris A. Modifying the diet and gut microbiota to
prevent and manage neurodegenerative diseases.

 Answer generated using hybrid RAG pipeline.

================================================================================
STEP 8: Clinical Verification
================================================================================
 Loading ClinicalBERT...
 ClinicalBERT ready!

VERIFICATION REPORT:
------------------------------------------------------------
Status           : MEDICALLY GROUNDED
Avg Similarity   : 0.9587
Top Chunk Scores : [0.963, 0.955, 0.957]
Answer Relevancy : 0.8974
```

BioResearch AI
Biomedical Research Assistant

New C

What is the recent findings of gut microbiome in neurodegenerative diseases from papers published in 2018-2025?

Recent findings suggest a strong association between gut microbiome dysbiosis and the pathogenesis of various neurodegenerative disorders, such as Parkinson's disease (PD), Alzheimer's disease (AD), and multiple sclerosis (MS). This association is thought to occur through the microbiota-gut brain axis (MGBA), which plays a crucial role in the development and progression of neurodegenerative diseases.

Copy

N

What is the recent findings of gut microbiome in neurodegenerative diseases from papers

Assumptions

- The system assumes that sufficient high-quality biomedical research articles are available from trusted open-access sources for retrieval
- It is assumed that users provide domain-relevant questions and the queries are in user query format.
- The system assumes uninterrupted access to literature repositories or local databases during retrieval.
- It is assumed that API calls used to retrieve papers are secured from threats like unauthorised access.

Workbreakdown and Responsiblities

Abhay P: Handled user query processing and retrieved relevant documents

Abel T John: Performed document parsing and generated embeddings

Achyuth Pradeep: Implemented answer generation using the Qwen2 model and integrated it into the pipeline.

Albin Sam Alex: Verified generated answers using ClinicalBERT and integrated the validation module into the system.

Hardware and Software Requirements

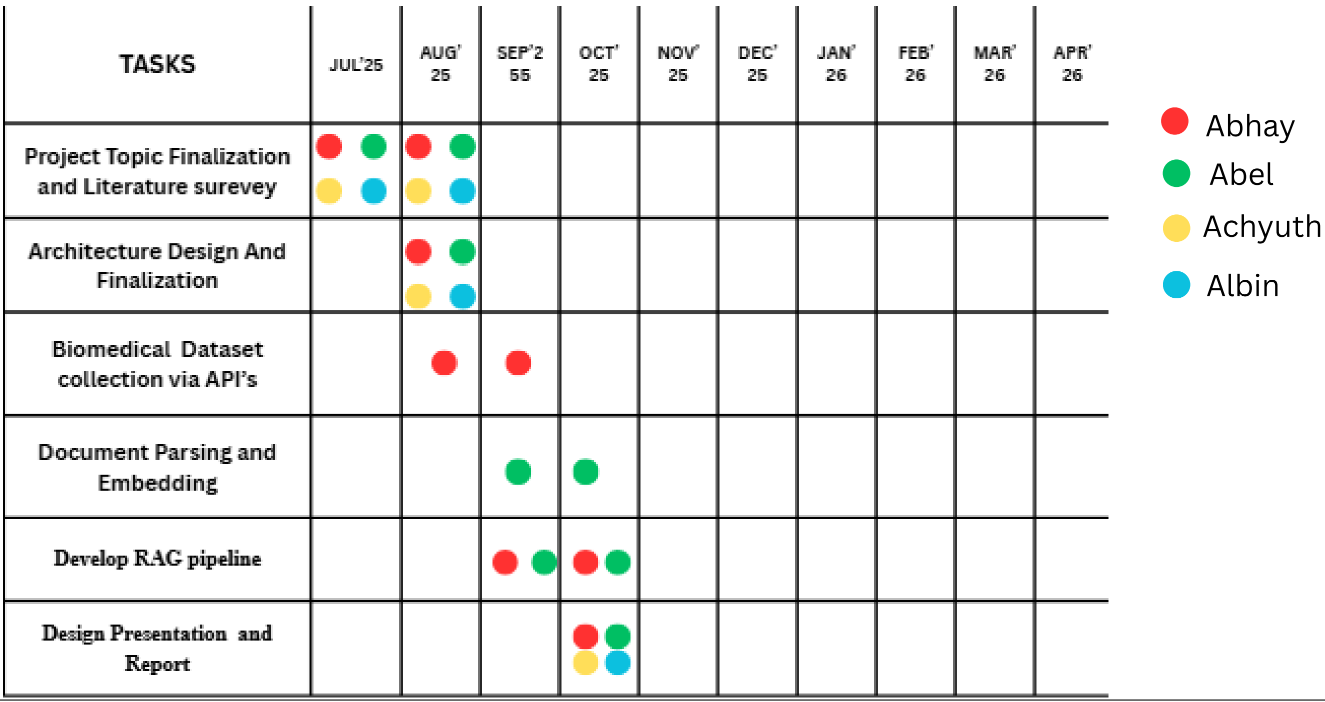
Software Stack:

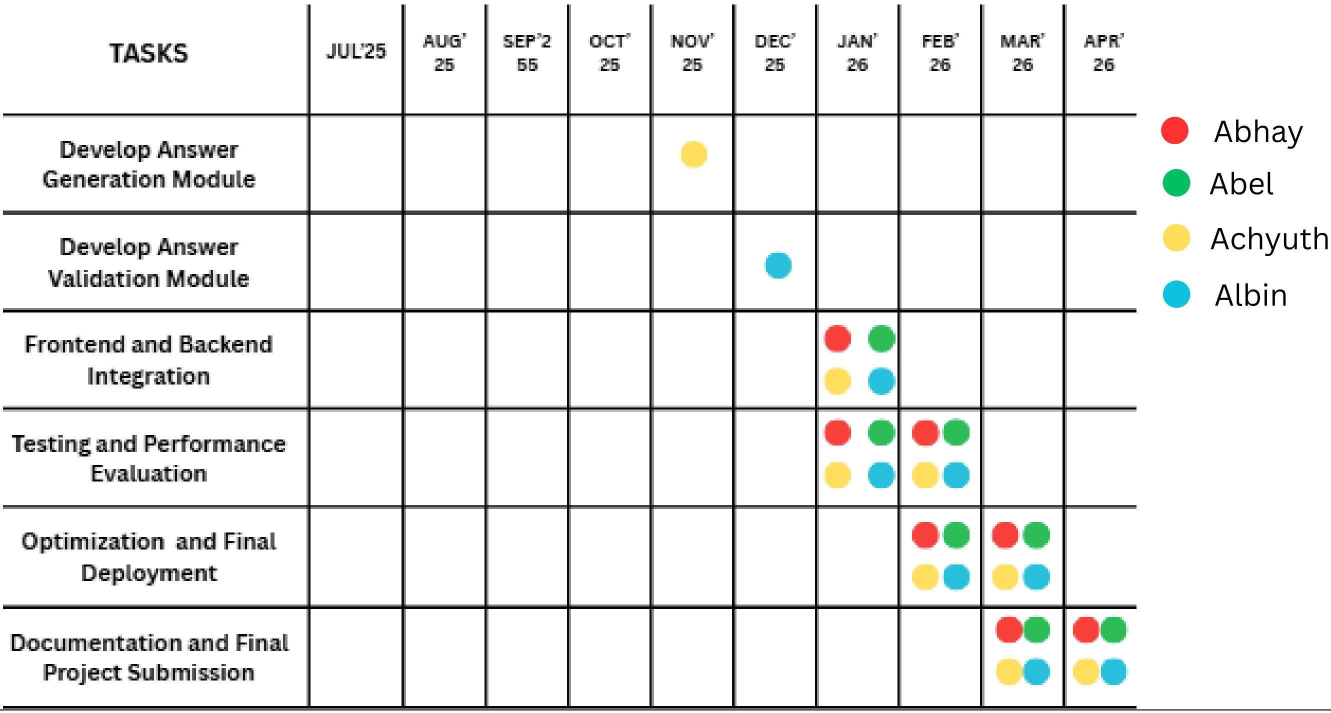
1) Embedding- NomicAI
2) Parsing-PyMuPDF
3) LLM- Qwen2
4) RealTime- EuropePMC
5) Filtering- PubmedBERT/ MINILM
6) Validation- ClinicalBERT

Hardware Requirements:

1) 8 GB RAM minimum
2) Stable internet connection (for API access)
3) Optional GPU (for embedding and faster processing)

Gantt Chart





Conclusion

- A modular Retrieval-Augmented Generation (RAG) pipeline was developed to separate document retrieval, chunk-level search, and answer generation, improving the overall reliability and scalability of the system.
- The system retrieves relevant biomedical research literature and generates evidence-supported answers with references, enhancing transparency and trust in the responses.
- Interpretable confidence scores are incorporated to evaluate how well the generated answer is grounded in retrieved documents and how relevant it is to the user's query.
- The lightweight generation module along with the retrieval strategy combined improves the relevance of retrieved information while maintaining computational efficiency and accurate answer generation.

Reference

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Thank You

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision
To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission
To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission
To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)
Engineering Graduates will be able to:

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2.Problem analysis:Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

5. Engineering Tool Usage:Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The engineer and the World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics:Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural,language, and learning differences.

10. Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning:Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	CO1: Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

Table 5.1: Course Outcomes

CO-PO and CO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 5.2: CO-PO-PSO Mapping

JUSTIFICATIONS FOR CO–PO MAPPING

Mapping	Low/Medium/High	Justification
101003/CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics and computing fundamentals to model and solve real world problems.

Mapping	Low/Medium/High	Justification
101003/CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental or sustainability aspects of engineering applications.
101003/CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real world problems.
101003/CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	Low/Medium/High	Justification
101003/CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.

Mapping	Low/Medium/High	Justification
101003/CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.

Mapping	**Low/Medium/High**	**Justification**
101003/CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.